In[ ]:=
```
n1 = 6;
g1 = 3;
n2 = 3;
g2 = 2;
n3 = 3;
g3 = 2;


(*Consistencies solution for the choice of pentaquark *)

γ12bar = 0.505242178637937;
β12 = 0.21321181216732096;
β22bar = 0.7946880881085071;

(*Defining Constants*)

xx = 0.346;
(*This is the value of αs which is αs=g^2/(ħc),
used from the mesonic fit as done in meson calculations.*)
alfas = xx;
(*aa=.178;*)
 zz = 306.;(*This is the mass of light quark like up, down*)
 aa = 0.197; (*This is the ratio of light mass to the heavy mass of quarks *)
yy = 107.6;
B0 = yy; (*Bag Constant *)
a1 = 1 / zz; (*This is α=ħ/(m1 c), where m1 is the mass of light quark. here,
reduced planck's constant and c both are set to be 1.*)
P = N[Pi, 7];
alfa1 = 1.;(* Ratio of mass of light quarks, BA, 13 September *)
alfa2 = aa; (* Ratio of mass of light quark to the heavy quark,
BA, 13 September *)

η = (n1 * g1 + n3 * g3 + n2 * g2) / 5;
theta = -6 η / (5 (5 η - 1)); (*Just a multiplication factor for a particular value of eta *)
Q1 = -0.5275862068965518; (*Q1 is the constant in the differential equation  d^2 f1/dx^2 =
```

Q1 $\frac{f1^{\frac{3}{2}}}{\sqrt{x}}$ for the outer region*)

```
Q3 = -1.357809759306383;
(*Q3 is the constant in the differential equation  d^2 f3/dx^2 =
```

Q3 $\frac{f3^{\frac{3}{2}}}{\sqrt{x}}$ in the middle region. This is obtained by solving two

consistency conditions along with other constant $\gamma$12bar. *)

```
Q2 = -3.6401613735622833;
(*Qbar is the constant in the differential equation  d^2 f3/dx^2 =
```

Q2 $\frac{f3^{\frac{3}{2}}}{\sqrt{x}}$ in the inner region. This is obtained by solving three

consistency conditions along with other constants $\beta$13 & $\beta$23. *)

```
g = alfas;(*Why  g = alfas in this case ?*)
(* I think there should be g = /sqrt{alfas} in order to satisfy the multiplication
  facor in the expreesion of kinetic energy i.e. 'c', BA, 13 September *)
```

$$R = \frac{3\,a}{(8 * alfas)} * \left(\frac{3\,\pi\,\eta}{2}\right)^{\frac{2}{3}};$$(*Page number 54, it is inverse in the copy though *)

$$c = \frac{12}{5\,\pi}\,\frac{4}{3}\,g^2 - \frac{4}{3}\,\frac{alfas}{a}\left(\frac{3\,\pi\,\eta}{2}\right)^{\frac{1}{3}};$$

(*Multiplication factor in the expression of kinetic energy *)

$$d = \frac{4}{3}\,g^2 - \frac{4}{3}\,\frac{alfas}{a}\left(\frac{3\,\pi\,\eta}{2}\right)^{\frac{1}{3}}\,\frac{18}{25\,\pi}\,\frac{\eta}{\left(\eta - \frac{1}{5}\right)};\quad (* \;\frac{18g^2}{5(5\eta\,-1\,)}.16\;\pi^2.\left(\frac{3a}{8alfas}\right).\;\frac{R^2}{(6\pi^2)^2}\;,$$

The value of R is from page number 54 which is written inverse here,

and $\left(\frac{3a}{8alfas}\right)$ is inverse in copy, BA, 13 September *)

(*Multiplication factor in the expression of Potential energy *)

(*Let us define the three functions for three regions *)

```
f3[x3_?NumericQ, x2_?NumericQ, u2_?NumericQ , s2_?NumericQ] :=
```

$$Abs\left[x3 * y3'[x3] - y3[x3] - n3 * \frac{Q3}{3\,\eta} - \beta 22bar * n2 * \frac{Q2}{3\,\eta} + \beta 22bar^{\frac{3}{2}} * n2 * \frac{Q3}{3\,\eta}\right]$$

```
(*Appendix from Kinetic energy*) /.
 First@NDSolve[{y3''[x] == (Q3 * Abs[y3[x]^(3/2)] / x^(1/2)), y3'[x2] == β22bar * s2 ,
    y3[x2] == β22bar * u2 (* It shows how the function f3
     (x3) changes on other regions*)}, y3, {x, x2, x3}];
(*This is the function f3 defined on the middle region. It has
  the differential equation  d^2 f3/dx^2 =
 Q3 f3^(3/2)/Sqrt[x] in for x3<x<x2 which is middle region. On this region, f1(x)=γ12 f2(x).  *)
f1[x1_?NumericQ, x3_?NumericQ, s3_?NumericQ, u3_?NumericQ] :=
  Abs[x1 * y1'[x1] - y1[x1] - n1 * Q1/(3 η) - γ12bar * n3 * Q3/(3 η) +
     γ12bar^(3/2) * n3 * Q1/(3 η) - (γ12bar^(3/2) * β22bar^(3/2) - β12^(3/2)) * n2 * Q1/(3 η) +
     γ12bar * β22bar^(3/2) * n2 * Q3/(3 η) - γ12bar * β22bar * n2 * Q2/(3 η)]
   (*Appendix from Kinetic energy*) /. First@NDSolve[{y1''[x] == (Q1 * Abs[y1[x]^(3/2)] / x^(1/2)),
     y1'[x3] == γ12bar * s3, (* It means at the boundary
      x3 the function df1 changes to other function *)
     y1[x3] == γ12bar * u3(*It means at the boundary x3 the
      function f1 changes to other function*)
    }, (* It means at the boundary x3 the function f1 changes to other function *)
    y1, {x, x3, x1}]; (*This is the function defined in outer region for x2<
  x<x1 so called f1. It has differential equation  d^2 f1/dx^2 =
 Q1 f1^(3/2)/Sqrt[x] for the outer region which is solved by using boundary
   condition of (d/dx) f1(x)|x2 = γ12bar (d/dx) f2(x)|x2 and f1(x3)=γ12 f2(x3)  *)
f2[xmax_?NumericQ, m2_?NumericQ] :=
```

```
  Module[{xx = xmax},
   solution2 = NDSolve[
     {y2''[x] == (Q2 * Abs[y2[x]^(3/2)] / x^(1/2)), y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, xx}];
   m = NIntegrate[Sqrt[x] * Evaluate[y2[x] /. solution2]^(3 / 2),
     {x, 1.*^-9, xx}, Method → {Automatic, "SymbolicProcessing" → 0}];
   z = m[[1]];
   z = z - n2 / (3. * η);
   (*z=z-n/(A)*);
   Return[Abs[z]]; Remove["NDSolve`*$*$1"]];
(*This is the function f3 defined on the inner region. It has the differential
  equation  d^2 f3/dx^2 = Q3 f3^(3/2)/Sqrt[x]  for 0<x<x_3 inner region. On this region,
f_1(x)=β_13 f_3(x) and f_2(x)= β_23 f_3(x).   *)

(*Start the program*)

m2 = 30;
For[j = 1., j ≤ 1., j++, B = (B0)^4;
  a = a1; (* It must be to satisfy the multiplication
   factor used in kinetic energy i.e. c, BA, 13 September *)
  (*Now for the first pentaquark,
  eta=1 which in total contains 5 quarks which can be written by *)

 mqn = If[j > 1, m2 + 2, m2]; (*case2new*)(*mqn=If[j>1,m2+4,m2];*)
  m2 = mqn;
  topi = 0.01;
  m2B = m2;
  Energy = 10^9;
  Energy = 10^9;
  Energy2plus = 10^9;
  H = 10^9 + 1;
  SN = 0;
```

```
nloop = 1;
For[n = 1, n ≤ 3, n++, While[Energy ≤ H, F = H;
  H = Energy;
  HX = Energy6bare;
  HY = Energy2plus; m2 = m2 + SN * 10^(-n);
  x2 = 2.*^-8; (*Print[m2];*)(*was 0 .5 1.66*)
  FindMinimum[f2[x2, m2], {x2, topi}];
  (*remove this function and investigate if it is actually required BS 4/3/2021*)
  x2 = x2 /. Last[FindMinimum[f2[x2, m2], {x2, topi}]];
  solution2 = NDSolve[{y2''[x] == (Q2 * Abs[y2[x]^(3/2)] / x^(1/2)),
     y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, x2}];
  t1 = Evaluate[y2[x2] /. solution2];
  t4 = Evaluate[y2'[x2] /. solution2];
  u2 = t1[[1]];
  s2 = t4[[1]];
  If[u2 < 0., u2 = 0.];

  x3 = x2 + 0.1;
  FindMinimum[{f3[x3, x2, u2, s2]}, {x3, x2}];
  x3 = x3 /. Last[FindMinimum[{f3[x3, x2, u2, s2]}, {x3, x2}]];
  solution3 = NDSolve[{y3''[x] == (Q3 * Abs[y3[x]^(3/2)] / x^(1/2)),
     y3'[x2] == β22bar * s2 , y3[x2] == β22bar * u2}, y3, {x, x2, x3}];
  t2 = Evaluate[y3[x3] /. solution3];
  t5 = Evaluate[y3'[x3] /. solution3];
  u3 = t2[[1]];
  s3 = t5[[1]];
  If[u3 < 0., u3 = 0.];

  x1 = x3 + 0.1;
  FindMinimum[{f1[x1, x3, s3, u3]}, {x1, x3}];
  x1 = x1 /. Last[FindMinimum[{f1[x1, x3, s3, u3]}, {x1, x3}]];
```

$$\text{solution1} = \text{NDSolve}\left[\left\{y1''[x] == \left(Q1 * \frac{\text{Abs}\left[y1[x]^{\frac{3}{2}}\right]}{x^{\frac{1}{2}}}\right),\right.\right.$$

$$\left.\left. y1'[x3] == \gamma 12\text{bar} * s3 \, , \, y1[x3] == \gamma 12\text{bar} * u3 \right\}, y1, \{x, x3, x1\}\right];$$

```
t3 = Evaluate[y1[x1] /. solution1];
t6 = Evaluate[y1'[x1] /. solution1];
u1 = t3[[1]];
s1 = t6[[1]];
If[u1 < 0., u1 = 0.];


(*Integrations *)(* These integrations change values but outer and outer 1
 doesnot because they depend on x and derivates of x (BA, 11 September)*)
```

$$\text{normmx2} = x2\, s2 - u2 \,;$$

$$\text{normx2} = \frac{n2\, Q2}{3\, \eta} \,;$$

```
(*normx2barr and normx2bar should be equal *)
```

$$\text{normmx3} = x3\, s3 - u3 \,;$$

$$\text{normx3} = \frac{n3\, Q3}{3\, \eta} + \beta 22\text{bar} \frac{n2\, Q2}{3\, \eta} - (\beta 22\text{bar})^{\frac{3}{2}} \frac{n2\, Q3}{3\, \eta};$$

```
(*normmx2 and normx2 should be equal *)
```

$$\text{normmx1} = x1\, s1 - u1 \,;$$

$$\text{normx1} = \frac{n1\, Q1}{3\, \eta} + \gamma 12\text{bar} \frac{n3\, Q3}{3\, \eta} - (\gamma 12\text{bar})^{\frac{3}{2}} \frac{n3\, Q1}{3\, \eta} +$$

$$\left((\gamma 12\text{bar})^{\frac{3}{2}} (\beta 22\text{bar})^{\frac{3}{2}} - (\beta 12)^{\frac{3}{2}}\right) \frac{n2\, Q1}{3\, \eta} - \gamma 12\text{bar} (\beta 22\text{bar})^{\frac{3}{2}} \frac{n2\, Q3}{3\, \eta} + \gamma 12\text{bar}\, \beta 22\text{bar} \frac{n2\, Q2}{3\, \eta} \,;$$

```
(*normmx1 and normx1 should be equal *)


h1 = NIntegrate[Evaluate[y2[x] /. solution2]^2.5 / (x^0.5),
  {x, 0, x2}, Method → {Automatic, "SymbolicProcessing" → 0}];
h2 = NIntegrate[Evaluate[y3[x] /. solution3]^2.5 / (x^0.5), {x, x2, x3},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h3 = NIntegrate[Evaluate[y1[x] /. solution1]^2.5 / (x^0.5), {x, x3, x1},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h4 = h1[[1]];
h5 = h2[[1]];
```

```
h6 = h3[[1]] ;
AB = 4/7 Sqrt[x2] (u2)^(5/2) - 5 n2/(21 η) s2 ;
CD = 4/7 Sqrt[x3] (u3)^(5/2) - 4/7 (β22bar)^(5/2) Sqrt[x2] (u2)^(5/2) -
   5/(7 Q3) s3 (n3 Q3/(3 η) + β22bar n2 Q2/(3 η) - (β22bar)^(3/2) n2 Q3/(3 η)) + 5/(7 Q3) (β22bar)^2 n2 Q2/(3 η) s2 ;
CCD = 4/7 Sqrt[x3] (u3)^(5/2) - 4/7 Sqrt[x2] (β22bar)^(5/2) (u2)^(5/2) - 5/(7 Q3) s3 normmx3 +
   5/(7 Q3) (β22bar)^2 normmx2 s2 ;

EF = 4/7 Sqrt[x1] (u1)^(5/2) - 4/7 (γ12bar)^(5/2) Sqrt[x3] (u3)^(5/2) -
   5/(7 Q1) s1 (n1 Q1/(3 η) + γ12bar n3 Q3/(3 η) - (γ12bar)^(3/2) n3 Q1/(3 η) + n2 Q1/(3 η) ((γ12bar)^(3/2) (β22bar)^(3/2) - (β12)^(3/2)) -
      (γ12bar) (β22bar)^(3/2) n2 Q3/(3 η) + (γ12bar) (β22bar) n2 Q2/(3 η)) +
   5/(7 Q1) (γ12bar)^2 s3 (n3 Q3/(3 η) + β22bar n2 Q2/(3 η) - (β22bar)^(3/2) n2 Q3/(3 η)) ;
(* h4 and AB should be equal *)
(* h5 and CD should be equal *)
(* h6 and EF should be equal *)

K222 = n2/(3 η Q2) s2 - 1/Q2 h4 ;
K22 = 4 n2/(7 η Q2) s2 - 4/(7 Q2) Sqrt[x2] (u2)^(5/2) ;
(*K2bar2barr and K2bar2bar should be equal *)
K32 = - 4/(7 Q2) β22bar^1.5 Sqrt[x2] (u2)^(5/2) +
```

$$\frac{n2}{3\,\eta}\left(\frac{\beta 22bar^{1.5}}{Q2}-\frac{\beta 22bar}{Q3}+\frac{5*\beta 22bar^{1.5}}{7\,Q2}\right)s2+\frac{n2}{3\,\eta\,Q3}s3\,;$$

$$K332=\frac{n2\,(\beta 22bar)^{\frac{3}{2}}}{3\,\eta\,Q2}s2-\frac{n2\,\beta 22bar}{3\,\eta\,Q3}s2+\frac{n2}{3\,\eta\,Q3}s3-\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2}h4;$$

(*K2bar2 and K2barr2 should be equal *)

$$K21=-\frac{4}{7\,Q2}\beta 12^{1.5}\sqrt{x2}\;(u2)^{\frac{5}{2}}+\frac{n2}{3\,\eta}\left(\frac{5}{7\,Q2}\beta 12^{1.5}+\frac{\beta 12^{1.5}}{Q2}-\frac{\gamma 12bar^{1.5}*\beta 22bar}{Q3}\right)s2+$$
$$\frac{n2}{3\,\eta}\left(\frac{\gamma 12bar^{1.5}}{Q3}-\frac{\gamma 12bar}{Q1}\right)s3+\frac{n2}{3\,\eta\,Q1}s1\,;$$

$$K221=\frac{n2}{3\,\eta\,Q2}(\beta 12)^{\frac{3}{2}}s2+\frac{n2\,(\gamma 12bar)^{\frac{3}{2}}}{3\,\eta\,Q3}s3-$$
$$\frac{n2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s2+\frac{n2}{3\,\eta\,Q1}s1-\frac{n2}{3\,\eta\,Q1}\gamma 12bar\,s3-\frac{(\beta 12)^{\frac{3}{2}}}{Q2}h4\,;$$

(*K2bar1 and K2barr1 should be equal *)

$$K31=\frac{4}{7}\beta 22bar^{1.5}\left(\frac{\gamma 12bar^{1.5}*\beta 22bar}{Q3}-\frac{\beta 12^{1.5}}{Q2}\right)\sqrt{x2}\;(u2)^{\frac{5}{2}}-\frac{4}{7\,Q3}\gamma 12bar^{1.5}\sqrt{x3}\;(u3)^{\frac{5}{2}}+$$
$$\frac{n2}{3\,\eta}\beta 22bar\left(\frac{5}{7\,Q2}\beta 22bar^{\frac{1}{2}}*\beta 12^{1.5}-\frac{\gamma 12bar^{1.5}*\beta 22bar^{1.5}}{Q3}-\frac{5\,Q2}{7\,Q3^2}\gamma 12bar^{1.5}*\right.$$
$$\left.\beta 22bar+\frac{Q2}{Q3^2}\gamma 12bar^{1.5}*\beta 22bar+\frac{1}{Q2}\beta 12^{1.5}*\beta 22bar^{\frac{1}{2}}-\frac{\beta 12^{1.5}}{Q3}\right)s2+$$
$$\frac{1}{3\,\eta\,Q3}\left(n3*\gamma 12bar^{1.5}-\frac{n3\,Q3}{Q1}\gamma 12bar+n2*\beta 12^{1.5}-\frac{n2\,Q2}{Q3}\gamma 12bar^{1.5}*\beta 22bar+\right.$$
$$\left.\frac{5}{7}\gamma 12bar^{1.5}\left(n3+\frac{n2\,Q2}{Q3}\beta 22bar-n2*\beta 22bar^{1.5}\right)\right)s3+\frac{n3}{3\,\eta\,Q1}s1\,;$$

$$(*K122 = \frac{n2}{3\,\eta}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}\left[\frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] - \frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4 - \frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5$$

$$- \frac{n2\,Q2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\left[\frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] + \frac{n3}{3\,\eta\,Q1}s1 - \frac{n3}{3\,\eta\,Q1}\gamma 12bar\,s3$$

$$+ (\beta 12)^{\frac{3}{2}}\frac{n2}{3\,\eta}\left[\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2}s2 + \frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] +$$

$$\frac{n3}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}s3\ ;\ *)$$

$$K331 = \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}\beta 22bar\,s2 - \frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4 - \frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5 -$$

$$\frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s3 + \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^2\,s2 + \frac{n2}{3\,\eta}(\beta 22bar)^{\frac{3}{2}}$$

$$\left(\frac{\beta 12^{\frac{3}{2}}}{Q2} - \frac{(\gamma 12bar)^{\frac{3}{2}}\beta 22bar}{Q3}\right)s2 + \frac{n3}{3\,\eta}\gamma 12bar\left(\frac{\gamma 12bar^{\frac{1}{2}}}{Q3} - \frac{1}{Q1}\right)s3 + \frac{n3}{3\,\eta\,Q1}s1\ ;$$

(*K12 and K122 should be equal*)

$$K33 = \frac{4}{7}\beta 22bar^2\left(\frac{\beta 22bar^{\frac{1}{2}}}{Q3} - \frac{\beta 22bar}{Q2}\right)\sqrt{x2}\,(u2)^{\frac{5}{2}} -$$

$$\frac{4}{7\,Q3}\sqrt{x3}\,(u3)^{\frac{5}{2}} + \frac{n2}{3\,\eta}\beta 22bar^2\left(\beta 22bar\frac{12}{7\,Q2} + \frac{2}{7}\frac{Q2}{Q3^2} - \frac{2*\beta 22bar^{\frac{1}{2}}}{Q3}\right)s2 +$$

$$\left(\frac{12\,n3}{21\,\eta\,Q3} + \beta 22bar\frac{n2}{3\,\eta\,Q3}\left(\beta 22bar^{\frac{1}{2}} - \frac{2}{7}\frac{Q2}{Q3} - \frac{5}{7}\beta 22bar^{\frac{1}{2}}\right)\right)s3\ ;$$

$$K333 = \frac{n2\,Q2}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2} - \frac{\beta 22bar}{Q3}\right)s3 - \frac{n2\,Q2\,\beta 22bar}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2} - \frac{\beta 22bar}{Q3}\right)s2 -$$

$$\frac{(\beta 22bar)^3}{Q2}h4 - \frac{1}{Q3}h5 + (\beta 22bar)^3\frac{n2}{3\,\eta\,Q2}s2 - (\beta 22bar)^{\frac{5}{2}}\frac{n2}{3\,\eta\,Q3}s2 + \frac{n3}{3\,\eta\,Q3}s3\ ;$$

(*K22 and K222 should be equal *)

(*THese Y and Z are to check the value of integral of power $\frac{5}{2}$ in the outer region. Which proved to be false yet. I need to work on it extensively.*)

$$K11 = \frac{4}{7}\left(\frac{\beta22bar^{\frac{5}{2}} * \gamma12bar^3}{Q3} - \frac{\beta12^3}{Q2}\right)\sqrt{x2}\ (u2)^{\frac{5}{2}} + \frac{4}{7}\left(\frac{\gamma12bar^{\frac{5}{2}}}{Q1} - \frac{\gamma12bar^3}{Q3}\right)\sqrt{x3}\ (u3)^{\frac{5}{2}} -$$

$$\frac{4}{7\,Q1}\sqrt{x1}\ (u1)^{\frac{5}{2}} + \left(\frac{n2}{3\,\eta}\left(\frac{\beta12^3}{Q2} - \frac{\beta12^{\frac{3}{2}} * \gamma12bar^{\frac{3}{2}} * \beta22bar}{Q3} + \frac{Q2}{Q3^2}\gamma12bar^3 *\right.\right.$$

$$\left.\beta22bar^2 - \frac{\beta12^{1.5} * \gamma12bar^{1.5} * \beta22bar}{Q3}\right) +$$

$$\left.\frac{5\,n2}{21\,\eta}\left(\frac{\beta12^3}{Q2} - \frac{Q2}{Q3^2}\beta22bar^2 * \gamma12bar^3\right)\right)s2 +$$

$$\left(\frac{n2}{3\,\eta}\left(\frac{\beta12^{1.5} * \gamma12bar^{1.5}}{Q3} - \frac{\beta12^{1.5} * \gamma12bar}{Q1} - \frac{\beta22bar^{1.5} * \gamma12bar^3}{Q3} +\right.\right.$$

$$\frac{\beta22bar^{1.5} * \gamma12bar^{\frac{5}{2}}}{Q1} + \frac{\beta12^{1.5} * \gamma12bar^{1.5}}{Q3} - \frac{Q2}{Q3^2}\gamma12bar^3 * \beta22bar -$$

$$\frac{\beta12^{1.5} * \gamma12bar}{Q1} + \frac{\gamma12bar^{\frac{5}{2}} * \beta22bar^{1.5}}{Q1} + \frac{Q2}{Q1^2}\gamma12bar^2 * \beta22bar -$$

$$\left.\left.\frac{Q3}{Q1^2}\beta22bar^{1.5} * \gamma12bar^2\right) + \frac{n3}{3\,\eta}\left(\frac{\gamma12bar^3}{Q3} - \frac{2 * \gamma12bar^{\frac{5}{2}}}{Q1} + \frac{Q3}{Q1^2}\gamma12bar^2\right) +\right.$$

$$\frac{5}{21\,\eta}\left(\frac{n3}{Q3}\gamma12bar^3 + \frac{n2\ Q2}{Q3^2}\gamma12bar^3 * \beta22bar - \frac{n2}{Q3}\beta22bar^{1.5} * \gamma12bar^3 -\right.$$

$$\left.\left.\frac{n3\ Q3}{Q1^2}\gamma12bar^2 - \frac{n2\ Q2}{Q1^2}\gamma12bar^2 * \beta22bar + \frac{n2\ Q3}{Q1^2}\gamma12bar^2 * \beta22bar^{1.5}\right)\right) *$$

$$s3 + \left(\frac{n1}{3\,\eta\,Q1} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\gamma12bar + \frac{n3}{3\,\eta\,Q1}\gamma12bar^{1.5} + \frac{n2}{3\,\eta\ Q1}\right.$$

$$\left(\beta 12^{1.5} - \gamma 12bar^{1.5} * \beta 22bar^{1.5} - \frac{Q2}{Q1}\beta 22bar * \gamma 12bar + \frac{Q3}{Q1}\beta 22bar^{1.5} * \gamma 12bar\right) + \frac{5}{21\,\eta}\left(\frac{n1}{Q1} + \frac{n3\ Q3}{Q1^2}\gamma 12bar - \frac{n3}{Q1}\gamma 12bar^{1.5} + \frac{n2}{Q1}\gamma 12bar^{1.5} * \beta 22bar^{1.5} - \frac{n2}{Q1}\beta 12^{1.5} - \frac{n2\ Q3}{Q1^2}\beta 22bar^{1.5} * \gamma 12bar + \frac{n2\ Q2}{Q1^2}\gamma 12bar\ \beta 22bar\right)\Bigg) * s1;$$

$$K111 = \frac{-(\beta 12)^3}{Q2}h4 - \frac{(\gamma 12bar)^3}{Q3}h5 - \frac{1}{Q1}h6 + \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^3\,\beta 22bar\,s3 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s2 + \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^3(\beta 22bar)^2 s2 + \left(\frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}} + \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{3}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\gamma 12bar - \frac{n2}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}} - \frac{n2\,Q2}{3\,\eta\,Q1^2}\beta 22bar\,\gamma 12bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}(\beta 22bar)^{\frac{3}{2}}\gamma 12bar\right)s1 - \left(\frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}}\gamma 12bar + \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}(\gamma 12bar)^2 - \frac{n2}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}}(\beta 22bar)^{\frac{3}{2}} - \frac{n2\,Q2}{3\,\eta\,Q1^2}(\gamma 12bar)^2\,\beta 22bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^2\right)s3 + \frac{n2}{3\,\eta}\frac{(\beta 12)^3}{Q2}s2 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s2 + \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}}\gamma 12bar\,s3 + \frac{n3}{3\,\eta\,Q3}(\gamma 12bar)^3 s3 - \frac{n2}{3\,\eta\,Q3}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^3 s3 - \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}}s3 + \frac{n2}{3\,\eta\,Q1}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^{\frac{5}{2}}s3 + \frac{n1}{3\,\eta\,Q1}s1;$$

(*K11 and K111 should be equal *)

$$T1 = c\left(\frac{4}{7}\left(alfa1 * g1\left(\beta 12^{\frac{5}{2}} - \gamma 12bar^{\frac{5}{2}} * \beta 22bar^{\frac{5}{2}}\right) + alfa2 * g2\right)\sqrt{x2}\ (u2)^{\frac{5}{2}} +$$

$$\frac{4}{7}\,\text{alfa2}*\text{g3}\sqrt{\text{x3}}\,(\text{u3})^{\frac{5}{2}}+\frac{4}{7}\,\text{alfa1}*\text{g1}\sqrt{\text{x1}}\,(\text{u1})^{\frac{5}{2}}+$$

$$\frac{5\,\text{n2}}{21\,\eta}\left(\beta 22\text{bar}^{2}\,\frac{\text{Q2}}{\text{Q3}}\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}}+\text{alfa2}*\text{g3}\right)-\right.$$

$$\left.\text{alfa1}*\text{g1}*\beta 12^{\frac{5}{2}}-\text{alfa2}*\text{g3}*\beta 22\text{bar}^{\frac{5}{2}}-\text{alfa2}*\text{g2}\right)\text{s2}+$$

$$\frac{5}{21\,\eta}\left(\frac{\gamma 12\text{bar}^{2}*\text{alfa1}*\text{g1}}{\text{Q1}}\left(\text{n3}*\text{Q3}+\beta 22\text{bar}*\text{n2}*\text{Q2}-\beta 22\text{bar}^{1.5}*\text{n2}*\text{Q3}\right)-\right.$$

$$\left.\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}}+\text{alfa2}*\text{g3}\right)\left(\text{n3}+\beta 22\text{bar}\,\frac{\text{n2}*\text{Q2}}{\text{Q3}}-\beta 22\text{bar}^{1.5}*\text{n2}\right)\right)\text{s3}-$$

$$\frac{5\,\text{alfa1}\,\text{g1}}{21\,\eta}\left(\text{n1}+\gamma 12\text{bar}\,\frac{\text{n3}\,\text{Q3}}{\text{Q1}}+\gamma 12\text{bar}^{1.5}\left(\beta 22\text{bar}^{1.5}*\text{n2}-\text{n3}\right)-\right.$$

$$\left.\beta 12^{1.5}*\text{n2}-\gamma 12\text{bar}*\beta 22\text{bar}^{1.5}\,\frac{\text{n2}\,\text{Q3}}{\text{Q1}}+\gamma 12\text{bar}\,\beta 22\text{bar}\,\frac{\text{n2}\,\text{Q2}}{\text{Q1}}\right)\text{s1}\Bigg);$$

$$\text{V1}=-\text{d}\left(\frac{(\text{g1}\,\text{n1}-1)}{\text{n1}}\,\text{g1}\,\text{K11}+\frac{(\text{g3}\,\text{n3}-1)}{\text{n3}}\,\text{g3}\,\text{K33}+\right.$$

$$\left.\frac{(\text{g2}\,\text{n2}-1)}{\text{n2}}\,\text{g2}\,\text{K22}+2\,\text{g1}\,\text{g3}\,\text{K31}+\frac{4}{3}\,\text{g2}\,\text{g1}\,\text{K21}+\frac{4}{3}\,\text{g2}\,\text{g3}\,\text{K32}\right);$$

```
(*These T1 and V1 are obtained
 by integrating manually which is being verified.*)
Energy6bare = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B + n1 * g1 * 306 + n3 * g3 * 1553 + n2 * g2 * 1553;
Energy = Energy6bare;
Volumeenergy = 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Energywithoutmass = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Print["R=  ", R];
Print["normx2 (computationally)=  ", normmx2];
Print["normx2 (by hand)=  ", normx2];
Print["normx3 (computationally)=  ", normmx3];
Print["normx3 (by hand)=  ", normx3];
Print["normx1 (computationally)=  ", normmx1];
Print["normx1 (by hand)=  ", normx1];
Print["0 to x2 (h4 computationally)=  ", h4];
Print["0 to x2 (h4 by hand)=  ", AB];
```

```
 Print["x2 to x3 (h5 computationally)=  ", h5];
 Print["x2 to x3 (h5 by hand)=  ", CD];
 Print["x2 to x3 (h5 by handdddd)=  ", CCD];
 Print["x3 to x1 (h6 computationally)=  ", h6];
 Print["x3 to x1 (h6 by hand)=  ", EF];
 Print["K22  (computationally) = ", K222];
 Print["K22  (by hand) = ", K22];
 Print["K32  (computationally) = ", K332];
 Print["K32  (by hand) = ", K32];
 Print["K21  (computationally) = ", K221];
 Print["K21  (by hand) =", K21];
 Print["K31  (computationally) = ", K331];
 Print["K31  (by hand) = ", K31];
 Print["K33  (computationally) = ", K333];
 Print["K33  (by hand) = ", K33];
 Print["K11  (computationally) = ", K111];
 Print["K11  (by hand) = ", K11];
 Print["x2= ", x2];
 Print["x3= ", x3];
 Print["x1=  ", x1];
  Print["T by hand=", T1];
 Print["V by hand=", V1];
 Print["m2=", m2];
 Print["R=  ", R];
 Print["Energywithoutmass = ", Energywithoutmass];
 Print["Energy=  ", Energy];
 If[n == 1, If[nloop == 1, nloop = nloop + 1; SN = 1. * 10^(-1);
   Energy11 = Energy, If[nloop == 2, nloop = nloop + 1;
    H = 10^9 + 1;
    m2 = m2B;
    If[Energy - Energy11 < 0, SN = 1., SN = -1.]]]];
If[n < 3, m2 = m2 - SN * 10^(-n); Energy = F; H = F];
If[n == 3, Energy6bare = HX;
 Print["x1=", x1];
 Print["x3=", x3];
 Print["x2=", x2];
 Print["Kinetic Energy=", T1];
 Print["Potential Energy=", V1];
 Print["Volume Energy=", Volumeenergy];
]]];

(*Code for Plotting function *)
```

```
Plot1 = Plot[Evaluate[((y2[x] /.solution2) / x)^(3 / 2)],
  {x, .00001, x2}, (*%PlotRange→{{0,0.65},{0,79.5}},*)
  (*Solution 1 = f3(x3), solution3 = f3(x3), solution3 = f1(x1) *)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17]
    , StyleForm["(f2(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot2 = Plot[Evaluate[((β22bar * y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot3 = Plot[Evaluate[((y3[x] /.solution3) / x)^(3 / 2)], {x, x2, x3},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot4 = Plot[Evaluate[((β12 * y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot5 = Plot[Evaluate[((γ12bar * y3[x] /. solution3) / x)^(3 / 2)],
  {x, x2, x3}, (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot6 = Plot[Evaluate[((y1[x] /. solution1) / x)^(3 / 2)], {x, x3, x1},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Show[{Plot2, Plot3}]
Show[{Plot4, Plot5, Plot6}]
```

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

General: Further output of FindMinimum::lstol will be suppressed during this calculation.

```
R=  0.0328721919
normx2 (computationally)=  -0.606693419
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115589
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928917
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  4.78902657
0 to x2 (h4 by hand)=  4.78902621
x2 to x3 (h5 computationally)=  1.06672753
x2 to x3 (h5 by hand)=  1.06672884
x2 to x3 (h5 by handdddd)=  1.06672791
x3 to x1 (h6 computationally)=  3.1129806
x3 to x1 (h6 by hand)=  3.1129802
K22  (computationally) = 0.226511186
K22  (by hand) = 0.226511086
K32  (computationally) = 0.211809016
K32  (by hand) = 0.211808945
K21  (computationally) = 0.242340888
K21  (by hand) =0.242340878
K31  (computationally) = 0.240786117
K31  (by hand) = 0.240786457
K33  (computationally) = 0.201094717
K33  (by hand) = 0.201095633
K11  (computationally) = 0.416086861
K11  (by hand) = 0.416086223
x2= 0.148100971
x3= 0.167102422
x1=  0.315816799
T by hand=712.730307
V by hand=-177.72529
m2=30
R=  0.0328721919
Energywithoutmass = 1163.25049
Energy=  25307.2505
R=  0.0328721919
```

```
normx2 (computationally)=  -0.606693504
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115581
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928906
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  4.79065902
0 to x2 (h4 by hand)=  4.79065905
x2 to x3 (h5 computationally)=  1.06709739
x2 to x3 (h5 by hand)=  1.06709795
x2 to x3 (h5 by handdddd)=  1.06709777
x3 to x1 (h6 computationally)=  3.11410199
x3 to x1 (h6 by hand)=  3.11410151
K22  (computationally) = 0.226549741
K22  (by hand) = 0.226549751
K32  (computationally) = 0.21184505
K32  (by hand) = 0.211845057
K21  (computationally) = 0.242383254
K21  (by hand) =0.242383255
K31  (computationally) = 0.240827944
K31  (by hand) = 0.240828093
K33  (computationally) = 0.201128155
K33  (by hand) = 0.201128571
K11  (computationally) = 0.416160049
K11  (by hand) = 0.416159189
x2= 0.148075741
x3= 0.167073755
x1=  0.315760285
T by hand=712.98301
V by hand=-177.756053
m2=30.01
R=  0.0328721919
Energywithoutmass = 1163.13523
Energy=  25 307.1352
R=  0.0328721919
normx2 (computationally)=  -0.606693506
```

```
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115581
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928905
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  4.80534543
0 to x2 (h4 by hand)=  4.80534547
x2 to x3 (h5 computationally)=  1.07043723
x2 to x3 (h5 by hand)=  1.07043778
x2 to x3 (h5 by handdddd)=  1.07043761
x3 to x1 (h6 computationally)=  3.12419626
x3 to x1 (h6 by hand)=  3.12419576
K22  (computationally) = 0.226894596
K22  (by hand) = 0.226894608
K32  (computationally) = 0.212168287
K32  (by hand) = 0.212168296
K21  (computationally) = 0.242764236
K21  (by hand) =0.242764238
K31  (computationally) = 0.241206637
K31  (by hand) = 0.241206783
K33  (computationally) = 0.201435629
K33  (by hand) = 0.201436038
K11  (computationally) = 0.416821324
K11  (by hand) = 0.41682044
x2= 0.147849181
x3= 0.166816539
x1=  0.315253066
T by hand=715.258272
V by hand=-178.034542
m2=30.1
R=  0.0328721919
Energywithoutmass = 1162.11095
Energy=  25306.111
R=  0.0328721919
normx2 (computationally)=  -0.606693458
normx2 (by hand)=  -0.606693562
```

```
normx3 (computationally)=  -0.548115582
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928905
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  4.82166382
0 to x2 (h4 by hand)=  4.82166364
x2 to x3 (h5 computationally)=  1.07414907
x2 to x3 (h5 by hand)=  1.07415001
x2 to x3 (h5 by handdddd)=  1.07414945
x3 to x1 (h6 computationally)=  3.13541338
x3 to x1 (h6 by hand)=  3.13541288
K22  (computationally) = 0.22727704
K22  (by hand) = 0.22727699
K32  (computationally) = 0.212526818
K32  (by hand) = 0.212526782
K21  (computationally) = 0.243186875
K21  (by hand) =0.24318687
K31  (computationally) = 0.241626891
K31  (by hand) = 0.241627137
K33  (computationally) = 0.201777132
K33  (by hand) = 0.201777802
K11  (computationally) = 0.417554973
K11  (by hand) = 0.41755411
x2= 0.147598651
x3= 0.166532127
x1=  0.314692293
T by hand=717.786605
V by hand=-178.343567
m2=30.2
R=  0.0328721919
Energywithoutmass = 1161.00153
Energy=  25305.0015
R=  0.0328721919
normx2 (computationally)=  -0.606693485
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115609
```

```
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928917
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  4.83798338
0 to x2 (h4 by hand)=  4.83798332
x2 to x3 (h5 computationally)=  1.07786061
x2 to x3 (h5 by hand)=  1.07786161
x2 to x3 (h5 by handdddd)=  1.07786099
x3 to x1 (h6 computationally)=  3.14663179
x3 to x1 (h6 by hand)=  3.14663124
K22  (computationally) = 0.227659035
K22  (by hand) = 0.22765902
K32  (computationally) = 0.212884858
K32  (by hand) = 0.212884847
K21  (computationally) = 0.243608819
K21  (by hand) =0.243608817
K31  (computationally) = 0.242046327
K31  (by hand) = 0.242046591
K33  (computationally) = 0.202117789
K33  (by hand) = 0.202118519
K11  (computationally) = 0.418287315
K11  (by hand) = 0.418286373
x2= 0.147349388
x3= 0.166249149
x1=  0.314134441
T by hand=720.315156
V by hand=-178.652019
m2=30.3
R=  0.0328721919
Energywithoutmass = 1159.922
Energy=  25 303.922
R=  0.0328721919
normx2 (computationally)=  -0.606693431
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115596
normx3 (by hand)=  -0.54811561
```

```
normx1 (computationally)=  -0.434928916
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  4.85430287
0 to x2 (h4 by hand)=  4.85430256
x2 to x3 (h5 computationally)=  1.08157279
x2 to x3 (h5 by hand)=  1.0815741
x2 to x3 (h5 by handdddd)=  1.08157318
x3 to x1 (h6 computationally)=  3.1578517
x3 to x1 (h6 by hand)=  3.15785123
K22  (computationally) = 0.228040201
K22  (by hand) = 0.228040116
K32  (computationally) = 0.213242187
K32  (by hand) = 0.213242126
K21  (computationally) = 0.244030059
K21  (by hand) =0.244030051
K31  (computationally) = 0.242465157
K31  (by hand) = 0.242465497
K33  (computationally) = 0.202458053
K33  (by hand) = 0.202458975
K11  (computationally) = 0.419018728
K11  (by hand) = 0.419017957
x2= 0.147101367
x3= 0.165967597
x1=  0.313579492
T by hand=722.843991
V by hand=-178.960044
m2=30.4
R=  0.0328721919
Energywithoutmass = 1158.87195
Energy=  25302.872
R=  0.0328721919
normx2 (computationally)=  -0.606693505
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.5481156
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928907
```

```
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  4.87062394
0 to x2 (h4 by hand)=  4.87062398
x2 to x3 (h5 computationally)=  1.08528382
x2 to x3 (h5 by hand)=  1.08528457
x2 to x3 (h5 by handdddd)=  1.08528421
x3 to x1 (h6 computationally)=  3.16907259
x3 to x1 (h6 by hand)=  3.16907199
K22  (computationally) = 0.228421055
K22  (by hand) = 0.228421067
K32  (computationally) = 0.213599084
K32  (by hand) = 0.213599092
K21  (computationally) = 0.244450599
K21  (by hand) =0.2444506
K31  (computationally) = 0.242882963
K31  (by hand) = 0.242883163
K33  (computationally) = 0.202796935
K33  (by hand) = 0.202797494
K11  (computationally) = 0.419748241
K11  (by hand) = 0.419747163
x2= 0.146854597
x3= 0.165687445
x1=  0.313027404
T by hand=725.372926
V by hand=-179.267269
m2=30.5
R=  0.0328721919
Energywithoutmass = 1157.85113
Energy=  25 301.8511
R=  0.0328721919
normx2 (computationally)=  -0.606693506
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115599
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928914
normx1 (by hand)=  -0.434928918
```

```
0 to x2 (h4 computationally)=  4.88694499
0 to x2 (h4 by hand)=  4.88694503
x2 to x3 (h5 computationally)=  1.088996
x2 to x3 (h5 by hand)=  1.08899675
x2 to x3 (h5 by handdddd)=  1.08899639
x3 to x1 (h6 computationally)=  3.18029513
x3 to x1 (h6 by hand)=  3.1802946
K22  (computationally) = 0.228801104
K22  (by hand) = 0.228801116
K32  (computationally) = 0.21395531
K32  (by hand) = 0.213955319
K21  (computationally) = 0.244870457
K21  (by hand) =0.244870458
K31  (computationally) = 0.243300302
K31  (by hand) = 0.243300501
K33  (computationally) = 0.2031358
K33  (by hand) = 0.203136359
K11  (computationally) = 0.420477232
K11  (by hand) = 0.420476296
x2= 0.14660905
x3= 0.165408704
x1=  0.312478177
T by hand=727.902223
V by hand=-179.574235
m2=30.6
R=  0.0328721919
Energywithoutmass = 1156.85907
Energy=  25300.8591
R=  0.0328721919
normx2 (computationally)=  -0.606693506
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115591
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928907
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  4.90326661
```

```
0 to x2 (h4 by hand)=  4.90326665
x2 to x3 (h5 computationally)=  1.09270829
x2 to x3 (h5 by hand)=  1.09270895
x2 to x3 (h5 by handdddd)=  1.09270868
x3 to x1 (h6 computationally)=  3.19151881
x3 to x1 (h6 by hand)=  3.19151825
K22  (computationally) = 0.229180531
K22  (by hand) = 0.229180543
K32  (computationally) = 0.214310948
K32  (by hand) = 0.214310957
K21  (computationally) = 0.245289615
K21  (by hand) =0.245289617
K31  (computationally) = 0.243716915
K31  (by hand) = 0.243717092
K33  (computationally) = 0.203474018
K33  (by hand) = 0.203474516
K11  (computationally) = 0.421204686
K11  (by hand) = 0.421203688
x2= 0.146364723
x3= 0.165131351
x1=  0.311931769
T by hand=730.431705
V by hand=-179.880601
m2=30.7
R=  0.0328721919
Energywithoutmass = 1155.89548
Energy=  25299.8955
```

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

```
R=  0.0328721919
normx2 (computationally)=  -0.60669348
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115581
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928905
```

```
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  4.91958858
0 to x2 (h4 by hand)=  4.91958851
x2 to x3 (h5 computationally)=  1.09642112
x2 to x3 (h5 by hand)=  1.0964219
x2 to x3 (h5 by handdddd)=  1.09642152
x3 to x1 (h6 computationally)=  3.20274389
x3 to x1 (h6 by hand)=  3.20274337
K22  (computationally) = 0.229559277
K22  (by hand) = 0.229559255
K32  (computationally) = 0.214665976
K32  (by hand) = 0.214665961
K21  (computationally) = 0.245708091
K21  (by hand) =0.245708089
K31  (computationally) = 0.24413292
K31  (by hand) = 0.244133123
K33  (computationally) = 0.203811866
K33  (by hand) = 0.203812425
K11  (computationally) = 0.42193117
K11  (by hand) = 0.421930259
x2= 0.146121602
x3= 0.16485538
x1=  0.311388166
T by hand=732.961463
V by hand=-180.186547
m2=30.8
R=  0.0328721919
Energywithoutmass = 1154.96
Energy=  25298.96
R=  0.0328721919
normx2 (computationally)=  -0.606693505
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115581
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928905
normx1 (by hand)=  -0.434928918
```

```
0 to x2 (h4 computationally)=  4.93591151
0 to x2 (h4 by hand)=  4.93591155
x2 to x3 (h5 computationally)=  1.10013371
x2 to x3 (h5 by hand)=  1.10013429
x2 to x3 (h5 by handdddd)=  1.10013411
x3 to x1 (h6 computationally)=  3.2139702
x3 to x1 (h6 by hand)=  3.21396968
K22  (computationally) = 0.229937531
K22  (by hand) = 0.229937542
K32  (computationally) = 0.215020492
K32  (by hand) = 0.2150205
K21  (computationally) = 0.246125886
K21  (by hand) =0.246125888
K31  (computationally) = 0.244548135
K31  (by hand) = 0.244548289
K33  (computationally) = 0.204148888
K33  (by hand) = 0.20414932
K11  (computationally) = 0.422656326
K11  (by hand) = 0.422655391
x2= 0.145879685
x3= 0.164580772
x1=  0.310847337
T by hand=735.491423
V by hand=-180.491918
m2=30.9
R=  0.0328721919
Energywithoutmass = 1154.05236
Energy=  25 298.0524
```

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

General: Further output of FindMinimum::fmdig will be suppressed during this calculation.

```
R=  0.0328721919
normx2 (computationally)=  -0.606693483
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.54811558
normx3 (by hand)=  -0.54811561
```

```
normx1 (computationally)=  -0.434928905
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  4.95223463
0 to x2 (h4 by hand)=  4.95223457
x2 to x3 (h5 computationally)=  1.10384715
x2 to x3 (h5 by hand)=  1.10384791
x2 to x3 (h5 by handdddd)=  1.10384756
x3 to x1 (h6 computationally)=  3.22519779
x3 to x1 (h6 by hand)=  3.22519727
K22  (computationally) = 0.230315059
K22  (by hand) = 0.230315041
K32  (computationally) = 0.215374382
K32  (by hand) = 0.215374369
K21  (computationally) = 0.246542999
K21  (by hand) =0.246542997
K31  (computationally) = 0.244962813
K31  (by hand) = 0.244963013
K33  (computationally) = 0.204485734
K33  (by hand) = 0.204486284
K11  (computationally) = 0.423380353
K11  (by hand) = 0.423379433
x2= 0.145638953
x3= 0.164307526
x1=  0.310309264
T by hand=738.021644
V by hand=-180.796866
m2=31.
R=  0.0328721919
Energywithoutmass = 1153.17215
Energy=  25 297.1722
R=  0.0328721919
normx2 (computationally)=  -0.606693497
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115581
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928907
```

```
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  4.96855858
0 to x2 (h4 by hand)=  4.96855859
x2 to x3 (h5 computationally)=  1.10756036
x2 to x3 (h5 by hand)=  1.10756101
x2 to x3 (h5 by handdddd)=  1.10756077
x3 to x1 (h6 computationally)=  3.2364267
x3 to x1 (h6 by hand)=  3.23642619
K22  (computationally) = 0.230692066
K22  (by hand) = 0.230692067
K32  (computationally) = 0.215727741
K32  (by hand) = 0.215727741
K21  (computationally) = 0.24695944
K21  (by hand) =0.24695944
K31  (computationally) = 0.245376715
K31  (by hand) = 0.245376887
K33  (computationally) = 0.204821752
K33  (by hand) = 0.204822231
K11  (computationally) = 0.42410323
K11  (by hand) = 0.424102337
x2= 0.145399404
x3= 0.16403562
x1=  0.309773921
T by hand=740.552085
V by hand=-181.101279
m2=31.1
R=  0.0328721919
Energywithoutmass = 1152.31913
Energy=  25296.3191
R=  0.0328721919
normx2 (computationally)=  -0.606693506
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115583
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928913
normx1 (by hand)=  -0.434928918
```

```
0 to x2 (h4 computationally)=  4.98488305
0 to x2 (h4 by hand)=  4.98488309
x2 to x3 (h5 computationally)=  1.11127391
x2 to x3 (h5 by hand)=  1.11127451
x2 to x3 (h5 by handdddd)=  1.11127432
x3 to x1 (h6 computationally)=  3.24765694
x3 to x1 (h6 by hand)=  3.24765649
K22  (computationally) = 0.231068455
K22  (by hand) = 0.231068468
K32  (computationally) = 0.216080528
K32  (by hand) = 0.216080537
K21  (computationally) = 0.247375212
K21  (by hand) =0.247375213
K31  (computationally) = 0.245789974
K31  (by hand) = 0.245790134
K33  (computationally) = 0.205157294
K33  (by hand) = 0.205157743
K11  (computationally) = 0.424825066
K11  (by hand) = 0.424824278
x2= 0.145161025
x3= 0.163765049
x1=  0.309241288
T by hand=743.08278
V by hand=-181.405242
m2=31.2
R=  0.0328721919
Energywithoutmass = 1151.49294
Energy=  25 295.4929
R=  0.0328721919
normx2 (computationally)=  -0.606693505
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115593
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928905
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.00120798
```

```
0 to x2 (h4 by hand)=  5.00120802
x2 to x3 (h5 computationally)=  1.11498796
x2 to x3 (h5 by hand)=  1.11498867
x2 to x3 (h5 by handdddd)=  1.11498837
x3 to x1 (h6 computationally)=  3.25888807
x3 to x1 (h6 by hand)=  3.25888745
K22  (computationally) = 0.231444217
K22  (by hand) = 0.231444229
K32  (computationally) = 0.216432745
K32  (by hand) = 0.216432753
K21  (computationally) = 0.247790295
K21  (by hand) =0.247790296
K31  (computationally) = 0.246202612
K31  (by hand) = 0.246202802
K33  (computationally) = 0.205492474
K33  (by hand) = 0.205493007
K11  (computationally) = 0.425545089
K11  (by hand) = 0.425543985
x2= 0.144923806
x3= 0.163495805
x1=  0.308711331
T by hand=745.613627
V by hand=-181.708594
m2=31.3
R=  0.0328721919
Energywithoutmass = 1150.69327
Energy=  25 294.6933
R=  0.0328721919
normx2 (computationally)=  -0.606693469
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115598
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928914
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.01753318
0 to x2 (h4 by hand)=  5.01753305
```

```
x2 to x3 (h5 computationally)=  1.11870263
x2 to x3 (h5 by hand)=  1.11870369
x2 to x3 (h5 by handdddd)=  1.11870304
x3 to x1 (h6 computationally)=  3.27012087
x3 to x1 (h6 by hand)=  3.27012033
K22  (computationally) = 0.23181929
K22  (by hand) = 0.231819253
K32  (computationally) = 0.216784354
K32  (by hand) = 0.216784327
K21  (computationally) = 0.248204729
K21  (by hand) =0.248204725
K31  (computationally) = 0.246614693
K31  (by hand) = 0.246614973
K33  (computationally) = 0.205827341
K33  (by hand) = 0.205828108
K11  (computationally) = 0.426264715
K11  (by hand) = 0.426263789
x2= 0.144687735
x3= 0.163227877
x1=  0.30818405
T by hand=748.14482
V by hand=-182.011673
m2=31.4
R=  0.0328721919
Energywithoutmass = 1149.9198
Energy=  25293.9198
R=  0.0328721919
normx2 (computationally)=  -0.606693506
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.5481156
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928912
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.0338595
0 to x2 (h4 by hand)=  5.03385955
x2 to x3 (h5 computationally)=  1.12241652
```

```
x2 to x3 (h5 by hand)=  1.12241731
x2 to x3 (h5 by handdddd)=  1.12241693
x3 to x1 (h6 computationally)=  3.28135472
x3 to x1 (h6 by hand)=  3.28135413
K22  (computationally) = 0.232193949
K22  (by hand) = 0.232193962
K32  (computationally) = 0.217135486
K32  (by hand) = 0.217135495
K21  (computationally) = 0.248618496
K21  (by hand) =0.248618497
K31  (computationally) = 0.247025874
K31  (by hand) = 0.247026086
K33  (computationally) = 0.206161057
K33  (by hand) = 0.206161648
K11  (computationally) = 0.426982729
K11  (by hand) = 0.426981685
x2= 0.144452812
x3= 0.16296124
x1=  0.3076594
T by hand=750.676142
V by hand=-182.314059
m2=31.5
R=  0.0328721919
Energywithoutmass = 1149.17231
Energy=  25293.1723
R=  0.0328721919
normx2 (computationally)=  -0.606693469
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.54811558
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928905
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.05018578
0 to x2 (h4 by hand)=  5.05018565
x2 to x3 (h5 computationally)=  1.12613123
x2 to x3 (h5 by hand)=  1.12613213
```

```
x2 to x3 (h5 by handdddd)=  1.12613165
x3 to x1 (h6 computationally)=  3.2925918
x3 to x1 (h6 by hand)=  3.29259237
K22  (computationally) = 0.23256783
K22  (by hand) = 0.232567794
K32  (computationally) = 0.217485958
K32  (by hand) = 0.217485932
K21  (computationally) = 0.249031691
K21  (by hand) =0.249031687
K31  (computationally) = 0.247436626
K31  (by hand) = 0.24743686
K33  (computationally) = 0.20649456
K33  (by hand) = 0.206495199
K11  (computationally) = 0.427703258
K11  (by hand) = 0.427704435
x2= 0.144219016
x3= 0.162695899
x1=  0.307137434
T by hand=753.208208
V by hand=-182.616864
m2=31.6
R=  0.0328721919
Energywithoutmass = 1148.45042
Energy=  25292.4504
R=  0.0328721919
normx2 (computationally)=  -0.606693483
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.54811558
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928918
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.066513
0 to x2 (h4 by hand)=  5.06651294
x2 to x3 (h5 computationally)=  1.12984585
x2 to x3 (h5 by hand)=  1.12984663
x2 to x3 (h5 by handdddd)=  1.12984627
```

```
x3 to x1 (h6 computationally)=  3.30382848
x3 to x1 (h6 by hand)=  3.30382921
K22  (computationally) = 0.232941244
K22  (by hand) = 0.232941227
K32  (computationally) = 0.217835949
K32  (by hand) = 0.217835937
K21  (computationally) = 0.249444156
K21  (by hand) =0.249444154
K31  (computationally) = 0.247846574
K31  (by hand) = 0.24784678
K33  (computationally) = 0.206827371
K33  (by hand) = 0.206827936
K11  (computationally) = 0.428419621
K11  (by hand) = 0.428421092
x2= 0.143986348
x3= 0.162431837
x1=  0.30661801
T by hand=755.74008
V by hand=-182.918462
m2=31.7
R=  0.0328721919
Energywithoutmass = 1147.75387
Energy=  25 291.7539
R=  0.0328721919
normx2 (computationally)=  -0.606693415
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.54811558
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928905
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.08284009
0 to x2 (h4 by hand)=  5.08283969
x2 to x3 (h5 computationally)=  1.13356181
x2 to x3 (h5 by hand)=  1.13356317
x2 to x3 (h5 by handdddd)=  1.13356223
x3 to x1 (h6 computationally)=  3.31506587
```

```
x3 to x1 (h6 by hand)=  3.31506645
K22  (computationally) = 0.233313862
K22  (by hand) = 0.233313752
K32  (computationally) = 0.218185292
K32  (by hand) = 0.218185214
K21  (computationally) = 0.249855933
K21  (by hand) =0.249855922
K31  (computationally) = 0.248256099
K31  (by hand) = 0.248256449
K33  (computationally) = 0.207160332
K33  (by hand) = 0.207161275
K11  (computationally) = 0.429133976
K11  (by hand) = 0.429135198
x2= 0.143754789
x3= 0.162169057
x1=  0.306101152
T by hand=758.272095
V by hand=-183.219488
m2=31.8
R=  0.0328721919
Energywithoutmass = 1147.08231
Energy=  25291.0823
R=  0.0328721919
normx2 (computationally)=  -0.606693504
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115596
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928905
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.09916899
0 to x2 (h4 by hand)=  5.09916903
x2 to x3 (h5 computationally)=  1.13727614
x2 to x3 (h5 by hand)=  1.13727692
x2 to x3 (h5 by handdddd)=  1.13727656
x3 to x1 (h6 computationally)=  3.3263046
x3 to x1 (h6 by hand)=  3.32630507
```

K22 (computationally) = 0.233686295

K22 (by hand) = 0.233686307

K32 (computationally) = 0.218534283

K32 (by hand) = 0.218534291

K21 (computationally) = 0.250267082

K21 (by hand) =0.250267083

K31 (computationally) = 0.248664563

K31 (by hand) = 0.24866477

K33 (computationally) = 0.207491668

K33 (by hand) = 0.207492246

K11 (computationally) = 0.429847275

K11 (by hand) = 0.429848252

x2= 0.143524348

x3= 0.16190752

x1= 0.305586847

T by hand=760.804284

V by hand=−183.519881

m2=31.9

R= 0.0328721919

Energywithoutmass = 1146.43561

Energy= 25 290.4356

R= 0.0328721919

normx2 (computationally)= −0.606693451

normx2 (by hand)= −0.606693562

normx3 (computationally)= −0.548115598

normx3 (by hand)= −0.54811561

normx1 (computationally)= −0.434928918

normx1 (by hand)= −0.434928918

0 to x2 (h4 computationally)= 5.11549725

0 to x2 (h4 by hand)= 5.11549702

x2 to x3 (h5 computationally)= 1.14099239

x2 to x3 (h5 by hand)= 1.14099364

x2 to x3 (h5 by handdddd)= 1.14099282

x3 to x1 (h6 computationally)= 3.33754495

x3 to x1 (h6 by hand)= 3.33754558

K22 (computationally) = 0.234057779

```
K22  (by hand) = 0.234057717
K32  (computationally) = 0.218882547
K32  (by hand) = 0.218882503
K21  (computationally) = 0.250677588
K21  (by hand) =0.250677582
K31  (computationally) = 0.249072786
K31  (by hand) = 0.249073112
K33  (computationally) = 0.207823488
K33  (by hand) = 0.207824377
K11  (computationally) = 0.43056029
K11  (by hand) = 0.430561597
x2= 0.143294993
x3= 0.16164725
x1=  0.305075091
T by hand=763.336867
V by hand=-183.820158
m2=32.
R=  0.0328721919
Energywithoutmass = 1145.81329
Energy=  25289.8133
R=  0.0328721919
normx2 (computationally)=  -0.606693488
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115581
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928905
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.13182679
0 to x2 (h4 by hand)=  5.13182675
x2 to x3 (h5 computationally)=  1.14470732
x2 to x3 (h5 by hand)=  1.14470809
x2 to x3 (h5 by handdddd)=  1.14470775
x3 to x1 (h6 computationally)=  3.3487862
x3 to x1 (h6 by hand)=  3.34878678
K22  (computationally) = 0.234428916
K22  (by hand) = 0.234428905
```

```
K32  (computationally) = 0.219230361
K32  (by hand) = 0.219230353
K21  (computationally) = 0.251087439
K21  (by hand) =0.251087438
K31  (computationally) = 0.249479993
K31  (by hand) = 0.249480195
K33  (computationally) = 0.208153813
K33  (by hand) = 0.208154371
K11  (computationally) = 0.431271428
K11  (by hand) = 0.431272598
x2= 0.143066734
x3= 0.161388203
x1=  0.304565835
T by hand=765.869509
V by hand=-184.119621
m2=32.1
R=  0.0328721919
Energywithoutmass = 1145.21527
Energy=  25289.2153
R=  0.0328721919
normx2 (computationally)=  -0.606693505
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.54811558
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928913
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.14815668
0 to x2 (h4 by hand)=  5.14815673
x2 to x3 (h5 computationally)=  1.14842301
x2 to x3 (h5 by hand)=  1.14842362
x2 to x3 (h5 by handdddd)=  1.14842344
x3 to x1 (h6 computationally)=  3.36002892
x3 to x1 (h6 by hand)=  3.36002962
K22  (computationally) = 0.234799424
K22  (by hand) = 0.234799437
K32  (computationally) = 0.219577621
```

```
K32  (by hand) = 0.21957763
K21  (computationally) = 0.251496662
K21  (by hand) =0.251496664
K31  (computationally) = 0.249886702
K31  (by hand) = 0.249886863
K33  (computationally) = 0.208483973
K33  (by hand) = 0.208484427
K11  (computationally) = 0.431982015
K11  (by hand) = 0.431983387
x2= 0.142839554
x3= 0.161130395
x1=  0.304059082
T by hand=768.402472
V by hand=-184.418811
m2=32.2
R=  0.0328721919
Energywithoutmass = 1144.64115
Energy=  25288.6411
R=  0.0328721919
normx2 (computationally)=  -0.606693415
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.54811561
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928907
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.16448621
0 to x2 (h4 by hand)=  5.1644858
x2 to x3 (h5 computationally)=  1.15214096
x2 to x3 (h5 by hand)=  1.15214266
x2 to x3 (h5 by handdddd)=  1.15214139
x3 to x1 (h6 computationally)=  3.37127224
x3 to x1 (h6 by hand)=  3.37127266
K22  (computationally) = 0.235169077
K22  (by hand) = 0.235168964
K32  (computationally) = 0.219924235
K32  (by hand) = 0.219924155
```

K21 (computationally) = 0.251905212

K21 (by hand) =0.251905201

K31 (computationally) = 0.250293217

K31 (by hand) = 0.250293659

K33 (computationally) = 0.208814917

K33 (by hand) = 0.208816116

K11 (computationally) = 0.43269057

K11 (by hand) = 0.432691527

x2= 0.142613436

x3= 0.160873829

x1= 0.303554792

T by hand=770.935606

V by hand=−184.717538

m2=32.3

R= 0.0328721919

Energywithoutmass = 1144.09058

Energy= 25 288.0906

R= 0.0328721919

normx2 (computationally)= −0.606693497

normx2 (by hand)= −0.606693562

normx3 (computationally)= −0.548115584

normx3 (by hand)= −0.54811561

normx1 (computationally)= −0.434928917

normx1 (by hand)= −0.434928918

0 to x2 (h4 computationally)= 5.18081769

0 to x2 (h4 by hand)= 5.1808177

x2 to x3 (h5 computationally)= 1.15585576

x2 to x3 (h5 by hand)= 1.15585649

x2 to x3 (h5 by handdddd)= 1.1558562

x3 to x1 (h6 computationally)= 3.38251769

x3 to x1 (h6 by hand)= 3.38251842

K22 (computationally) = 0.235538608

K22 (by hand) = 0.23553861

K32 (computationally) = 0.22027048

K32 (by hand) = 0.220270482

K21 (computationally) = 0.25231318

```
K21  (by hand) =0.25231318
K31  (computationally) = 0.250698364
K31  (by hand) = 0.250698556
K33  (computationally) = 0.209143196
K33  (by hand) = 0.20914373
K11  (computationally) = 0.433399369
K11  (by hand) = 0.433400827
x2= 0.142388396
x3= 0.160618438
x1=  0.303052969
T by hand=773.469013
V by hand=-185.01575
m2=32.4
R=  0.0328721919
Energywithoutmass = 1143.5636
Energy=  25 287.5636
R=  0.0328721919
normx2 (computationally)=  -0.606693504
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.5481156
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928905
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.19714906
0 to x2 (h4 by hand)=  5.1971491
x2 to x3 (h5 computationally)=  1.15957258
x2 to x3 (h5 by hand)=  1.15957342
x2 to x3 (h5 by handdddd)=  1.15957302
x3 to x1 (h6 computationally)=  3.3937634
x3 to x1 (h6 by hand)=  3.39376386
K22  (computationally) = 0.23590737
K22  (by hand) = 0.235907382
K32  (computationally) = 0.22061613
K32  (by hand) = 0.220616138
K21  (computationally) = 0.252720468
K21  (by hand) =0.252720469
```

K31 (computationally) = 0.251103254

K31 (by hand) = 0.251103478

K33 (computationally) = 0.209472132

K33 (by hand) = 0.209472757

K11 (computationally) = 0.43410556

K11 (by hand) = 0.434106509

x2= 0.142164404

x3= 0.160364271

x1= 0.302553561

T by hand=776.002504

V by hand=−185.313347

m2=32.5

R= 0.0328721919

Energywithoutmass = 1143.05968

Energy= 25 287.0597

R= 0.0328721919

normx2 (computationally)= −0.606693466

normx2 (by hand)= −0.606693562

normx3 (computationally)= −0.548115584

normx3 (by hand)= −0.54811561

normx1 (computationally)= −0.434928905

normx1 (by hand)= −0.434928918

0 to x2 (h4 computationally)= 5.21348057

0 to x2 (h4 by hand)= 5.21348042

x2 to x3 (h5 computationally)= 1.16328968

x2 to x3 (h5 by hand)= 1.16329069

x2 to x3 (h5 by handdddd)= 1.16329013

x3 to x1 (h6 computationally)= 3.40501092

x3 to x1 (h6 by hand)= 3.40501149

K22 (computationally) = 0.236275449

K22 (by hand) = 0.236275407

K32 (computationally) = 0.22096117

K32 (by hand) = 0.22096114

K21 (computationally) = 0.253127148

K21 (by hand) =0.253127144

K31 (computationally) = 0.251507556

```
K31  (by hand) = 0.25150782
K33  (computationally) = 0.209800543
K33  (by hand) = 0.209801264
K11  (computationally) = 0.434811736
K11  (by hand) = 0.434812913
x2= 0.141941456
x3= 0.160111293
x1=  0.302056583
T by hand=778.536353
V by hand=-185.610726
m2=32.6
R=  0.0328721919
Energywithoutmass = 1142.57862
Energy=  25 286.5786
R=  0.0328721919
normx2 (computationally)=  -0.606693467
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115581
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928913
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.22981293
0 to x2 (h4 by hand)=  5.22981279
x2 to x3 (h5 computationally)=  1.16700666
x2 to x3 (h5 by hand)=  1.16700762
x2 to x3 (h5 by handdddd)=  1.16700711
x3 to x1 (h6 computationally)=  3.41625967
x3 to x1 (h6 by hand)=  3.41626038
K22  (computationally) = 0.236643067
K22  (by hand) = 0.236643027
K32  (computationally) = 0.221305738
K32  (by hand) = 0.22130571
K21  (computationally) = 0.253533209
K21  (by hand) =0.253533205
K31  (computationally) = 0.251911162
K31  (by hand) = 0.251911413
```

```
K33  (computationally) = 0.210128265
K33  (by hand) = 0.21012895
K11  (computationally) = 0.4355169
K11  (by hand) = 0.435518333
x2= 0.141719549
x3= 0.159859498
x1=  0.301562
T by hand=781.070421
V by hand=-185.907639
m2=32.7
R=  0.0328721919
Energywithoutmass = 1142.12022
Energy=  25286.1202
R=  0.0328721919
normx2 (computationally)=  -0.606693492
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.54811558
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928907
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.24614599
0 to x2 (h4 by hand)=  5.24614597
x2 to x3 (h5 computationally)=  1.1707236
x2 to x3 (h5 by hand)=  1.17072434
x2 to x3 (h5 by handdddd)=  1.17072405
x3 to x1 (h6 computationally)=  3.42750927
x3 to x1 (h6 by hand)=  3.42750991
K22  (computationally) = 0.237010182
K22  (by hand) = 0.237010176
K32  (computationally) = 0.221649809
K32  (by hand) = 0.221649805
K21  (computationally) = 0.253938633
K21  (by hand) =0.253938633
K31  (computationally) = 0.252314072
K31  (by hand) = 0.252314268
K33  (computationally) = 0.210455326
```

```
K33  (by hand) = 0.21045587
K11  (computationally) = 0.436220363
K11  (by hand) = 0.436221652
x2= 0.141498675
x3= 0.159608874
x1=  0.301069784
T by hand=783.604609
V by hand=-186.203926
m2=32.8
R=  0.0328721919
Energywithoutmass = 1141.68422
Energy=  25285.6842
R=  0.0328721919
normx2 (computationally)=  -0.606693476
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115601
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928904
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.26247922
0 to x2 (h4 by hand)=  5.26247911
x2 to x3 (h5 computationally)=  1.17444172
x2 to x3 (h5 by hand)=  1.17444283
x2 to x3 (h5 by handdddd)=  1.17444217
x3 to x1 (h6 computationally)=  3.43875986
x3 to x1 (h6 by hand)=  3.43876031
K22  (computationally) = 0.237376633
K22  (by hand) = 0.237376605
K32  (computationally) = 0.221993323
K32  (by hand) = 0.221993304
K21  (computationally) = 0.254343427
K21  (by hand) =0.254343424
K31  (computationally) = 0.252716581
K31  (by hand) = 0.252716875
K33  (computationally) = 0.210782517
K33  (by hand) = 0.210783327
```

K11 (computationally) = 0.436922516

K11 (by hand) = 0.436923487

x2= 0.141278821

x3= 0.159359427

x1= 0.300579923

T by hand=786.139019

V by hand=−186.499814

m2=32.9

R= 0.0328721919

Energywithoutmass = 1141.27031

Energy= 25 285.2703

R= 0.0328721919

normx2 (computationally)= −0.606693504

normx2 (by hand)= −0.606693562

normx3 (computationally)= −0.548115595

normx3 (by hand)= −0.54811561

normx1 (computationally)= −0.434928913

normx1 (by hand)= −0.434928918

0 to x2 (h4 computationally)= 5.27881332

0 to x2 (h4 by hand)= 5.27881337

x2 to x3 (h5 computationally)= 1.17815893

x2 to x3 (h5 by hand)= 1.17815974

x2 to x3 (h5 by handdddd)= 1.17815939

x3 to x1 (h6 computationally)= 3.45001216

x3 to x1 (h6 by hand)= 3.45001277

K22 (computationally) = 0.237742643

K22 (by hand) = 0.237742656

K32 (computationally) = 0.222336349

K32 (by hand) = 0.222336358

K21 (computationally) = 0.25474763

K21 (by hand) =0.254747631

K31 (computationally) = 0.25311824

K31 (by hand) = 0.253118454

K33 (computationally) = 0.211108478

K33 (by hand) = 0.211109078

K11 (computationally) = 0.437624452

```
K11  (by hand) = 0.437625684
x2= 0.141059985
x3= 0.159111123
x1=  0.300092411
T by hand=788.673714
V by hand=-186.795324
m2=33.
R=  0.0328721919
Energywithoutmass = 1140.87834
Energy=  25 284.8783
R=  0.0328721919
normx2 (computationally)=  -0.606693463
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115599
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928918
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.29514734
0 to x2 (h4 by hand)=  5.29514718
x2 to x3 (h5 computationally)=  1.18187753
x2 to x3 (h5 by hand)=  1.18187874
x2 to x3 (h5 by handdddd)=  1.18187799
x3 to x1 (h6 computationally)=  3.46126544
x3 to x1 (h6 by hand)=  3.4612661
K22  (computationally) = 0.238107915
K22  (by hand) = 0.238107869
K32  (computationally) = 0.222678776
K32  (by hand) = 0.222678744
K21  (computationally) = 0.255151203
K21  (by hand) =0.255151198
K31  (computationally) = 0.253519545
K31  (by hand) = 0.253519863
K33  (computationally) = 0.21143467
K33  (by hand) = 0.21143554
K11  (computationally) = 0.438325102
K11  (by hand) = 0.438326454
```

```
x2= 0.14084215
x3= 0.158863978
x1=  0.299607218
T by hand=791.208634
V by hand=-187.090456
m2=33.1
R=  0.0328721919
Energywithoutmass = 1140.50797
Energy=  25284.508
R=  0.0328721919
normx2 (computationally)=  -0.606693504
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115592
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928905
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.31148257
0 to x2 (h4 by hand)=  5.31148261
x2 to x3 (h5 computationally)=  1.18559498
x2 to x3 (h5 by hand)=  1.18559575
x2 to x3 (h5 by handddddd)=  1.18559544
x3 to x1 (h6 computationally)=  3.47251964
x3 to x1 (h6 by hand)=  3.47252017
K22  (computationally) = 0.238472854
K22  (by hand) = 0.238472867
K32  (computationally) = 0.22302078
K32  (by hand) = 0.223020789
K21  (computationally) = 0.255554158
K21  (by hand) =0.255554159
K31  (computationally) = 0.253919916
K31  (by hand) = 0.253920122
K33  (computationally) = 0.211759517
K33  (by hand) = 0.211760094
K11  (computationally) = 0.439024089
K11  (by hand) = 0.439025184
x2= 0.140625322
```

```
x3= 0.158617958
x1=  0.299124316
T by hand=793.743631
V by hand=-187.384857
m2=33.2
R=  0.0328721919
Energywithoutmass = 1140.15911
Energy=  25 284.1591
R=  0.0328721919
normx2 (computationally)=  -0.606693422
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115585
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928916
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.32781724
0 to x2 (h4 by hand)=  5.32781686
x2 to x3 (h5 computationally)=  1.18931441
x2 to x3 (h5 by hand)=  1.18931584
x2 to x3 (h5 by handdddd)=  1.18931487
x3 to x1 (h6 computationally)=  3.48377548
x3 to x1 (h6 by hand)=  3.48377622
K22  (computationally) = 0.238836914
K22  (by hand) = 0.23883681
K32  (computationally) = 0.223362115
K32  (by hand) = 0.223362041
K21  (computationally) = 0.255956515
K21  (by hand) =0.255956505
K31  (computationally) = 0.254320107
K31  (by hand) = 0.254320478
K33  (computationally) = 0.21208495
K33  (by hand) = 0.212085951
K11  (computationally) = 0.439723077
K11  (by hand) = 0.439724608
x2= 0.140409475
x3= 0.158373085
```

```
x1=  0.298643716
T by hand=796.279047
V by hand=-187.679237
m2=33.3
R=  0.0328721919
Energywithoutmass = 1139.83132
Energy=  25 283.8313
R=  0.0328721919
normx2 (computationally)=  -0.606693435
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115594
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928907
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.34415321
0 to x2 (h4 by hand)=  5.34415289
x2 to x3 (h5 computationally)=  1.193033
x2 to x3 (h5 by hand)=  1.19303443
x2 to x3 (h5 by handdddd)=  1.19303346
x3 to x1 (h6 computationally)=  3.49503187
x3 to x1 (h6 by hand)=  3.49503243
K22  (computationally) = 0.23920068
K22  (by hand) = 0.239200592
K32  (computationally) = 0.223703068
K32  (by hand) = 0.223703006
K21  (computationally) = 0.256358249
K21  (by hand) =0.25635824
K31  (computationally) = 0.25471943
K31  (by hand) = 0.254719802
K33  (computationally) = 0.212409284
K33  (by hand) = 0.212410293
K11  (computationally) = 0.440419856
K11  (by hand) = 0.440421049
x2= 0.14019462
x3= 0.158129325
x1=  0.298165364
```

```
T by hand=798.81448
V by hand=-187.972808
m2=33.4
R=  0.0328721919
Energywithoutmass = 1139.52457
Energy=  25283.5246
R=  0.0328721919
normx2 (computationally)=  -0.606693439
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115586
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928917
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.3604896
0 to x2 (h4 by hand)=  5.36048931
x2 to x3 (h5 computationally)=  1.19675161
x2 to x3 (h5 by hand)=  1.19675293
x2 to x3 (h5 by handdddd)=  1.19675208
x3 to x1 (h6 computationally)=  3.50629001
x3 to x1 (h6 by hand)=  3.50629077
K22  (computationally) = 0.239563878
K22  (by hand) = 0.239563796
K32  (computationally) = 0.224043486
K32  (by hand) = 0.224043428
K21  (computationally) = 0.256759404
K21  (by hand) =0.256759396
K31  (computationally) = 0.255118134
K31  (by hand) = 0.255118476
K33  (computationally) = 0.212732984
K33  (by hand) = 0.212733911
K11  (computationally) = 0.441116639
K11  (by hand) = 0.441118193
x2= 0.139980742
x3= 0.157886674
x1=  0.297689274
T by hand=801.350255
```

V by hand=−188.266156

m2=33.5

R= 0.0328721919

Energywithoutmass = 1139.23855

Energy= 25 283.2385

R= 0.0328721919

normx2 (computationally)= −0.606693489

normx2 (by hand)= −0.606693562

normx3 (computationally)= −0.548115581

normx3 (by hand)= −0.54811561

normx1 (computationally)= −0.434928918

normx1 (by hand)= −0.434928918

0 to x2 (h4 computationally)= 5.37682689

0 to x2 (h4 by hand)= 5.37682686

x2 to x3 (h5 computationally)= 1.20046981

x2 to x3 (h5 by hand)= 1.20047062

x2 to x3 (h5 by handdddd)= 1.20047028

x3 to x1 (h6 computationally)= 3.51754904

x3 to x1 (h6 by hand)= 3.51754983

K22 (computationally) = 0.23992666

K22 (by hand) = 0.239926652

K32 (computationally) = 0.224383458

K32 (by hand) = 0.224383452

K21 (computationally) = 0.257159951

K21 (by hand) =0.25715995

K31 (computationally) = 0.255516086

K31 (by hand) = 0.2555163

K33 (computationally) = 0.213055824

K33 (by hand) = 0.213056417

K11 (computationally) = 0.441811873

K11 (by hand) = 0.441813458

x2= 0.139767836

x3= 0.157645123

x1= 0.297215408

T by hand=803.886147

V by hand=−188.558885

```
m2=33.6
R=  0.0328721919
Energywithoutmass = 1138.97309
Energy=  25 282.9731
R=  0.0328721919
normx2 (computationally)=  -0.606693463
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115592
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928918
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.39316401
0 to x2 (h4 by hand)=  5.39316384
x2 to x3 (h5 computationally)=  1.20418962
x2 to x3 (h5 by hand)=  1.20419079
x2 to x3 (h5 by handdddd)=  1.20419009
x3 to x1 (h6 computationally)=  3.52880901
x3 to x1 (h6 by hand)=  3.52880973
K22  (computationally) = 0.240288696
K22  (by hand) = 0.240288649
K32  (computationally) = 0.224722839
K32  (by hand) = 0.224722806
K21  (computationally) = 0.257559884
K21  (by hand) =0.25755988
K31  (computationally) = 0.255913757
K31  (by hand) = 0.255914064
K33  (computationally) = 0.213379066
K33  (by hand) = 0.213379906
K11  (computationally) = 0.442505867
K11  (by hand) = 0.442507342
x2= 0.139555887
x3= 0.157404682
x1=  0.296743754
T by hand=806.42227
V by hand=-188.85128
m2=33.7
```

```
R=  0.0328721919
Energywithoutmass = 1138.72784
Energy=  25 282.7278
R=  0.0328721919
normx2 (computationally)=  −0.606693434
normx2 (by hand)=  −0.606693562
normx3 (computationally)=  −0.548115594
normx3 (by hand)=  −0.54811561
normx1 (computationally)=  −0.434928907
normx1 (by hand)=  −0.434928918
0 to x2 (h4 computationally)=  5.4095016
0 to x2 (h4 by hand)=  5.40950127
x2 to x3 (h5 computationally)=  1.20790952
x2 to x3 (h5 by hand)=  1.20791097
x2 to x3 (h5 by handdddd)=  1.20790999
x3 to x1 (h6 computationally)=  3.54006998
x3 to x1 (h6 by hand)=  3.54007054
K22  (computationally) = 0.240650188
K22  (by hand) = 0.240650099
K32  (computationally) = 0.225061706
K32  (by hand) = 0.225061643
K21  (computationally) = 0.257959214
K21  (by hand) =0.257959205
K31  (computationally) = 0.256310798
K31  (by hand) = 0.256311176
K33  (computationally) = 0.213701733
K33  (by hand) = 0.21370276
K11  (computationally) = 0.443198546
K11  (by hand) = 0.443199745
x2= 0.139344893
x3= 0.157165326
x1=  0.296274294
T by hand=808.958554
V by hand=−189.143154
m2=33.8
R=  0.0328721919
```

```
Energywithoutmass = 1138.50268
Energy=  25 282.5027
R=  0.0328721919
normx2 (computationally)=  -0.606693493
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115579
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928907
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.42584042
0 to x2 (h4 by hand)=  5.42584042
x2 to x3 (h5 computationally)=  1.21162804
x2 to x3 (h5 by hand)=  1.2116288
x2 to x3 (h5 by handdddd)=  1.21162852
x3 to x1 (h6 computationally)=  3.55133253
x3 to x1 (h6 by hand)=  3.55133321
K22  (computationally) = 0.241011382
K22  (by hand) = 0.24101138
K32  (computationally) = 0.22540017
K32  (by hand) = 0.225400169
K21  (computationally) = 0.258357978
K21  (by hand) =0.258357978
K31  (computationally) = 0.256706905
K31  (by hand) = 0.256707105
K33  (computationally) = 0.214022927
K33  (by hand) = 0.214023484
K11  (computationally) = 0.443890868
K11  (by hand) = 0.443892238
x2= 0.139134853
x3= 0.156927035
x1=  0.295807019
T by hand=811.495075
V by hand=-189.434586
m2=33.9
R=  0.0328721919
Energywithoutmass = 1138.29747
```

```
Energy=  25282.2975
R=  0.0328721919
normx2 (computationally)=  -0.606693411
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.54811558
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928907
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.44217851
0 to x2 (h4 by hand)=  5.44217806
x2 to x3 (h5 computationally)=  1.21534913
x2 to x3 (h5 by hand)=  1.21535066
x2 to x3 (h5 by handdddd)=  1.21534961
x3 to x1 (h6 computationally)=  3.56259598
x3 to x1 (h6 by hand)=  3.56259666
K22  (computationally) = 0.24137166
K22  (by hand) = 0.241371537
K32  (computationally) = 0.225737967
K32  (by hand) = 0.22573788
K21  (computationally) = 0.25875613
K21  (by hand) =0.258756118
K31  (computationally) = 0.257102957
K31  (by hand) = 0.257103352
K33  (computationally) = 0.214345114
K33  (by hand) = 0.214346175
K11  (computationally) = 0.444582003
K11  (by hand) = 0.444583434
x2= 0.138925743
x3= 0.156689838
x1=  0.29534191
T by hand=814.031864
V by hand=-189.725794
m2=34.
R=  0.0328721919
Energywithoutmass = 1138.11178
Energy=  25282.1118
```

```
R=  0.0328721919
normx2 (computationally)=  -0.606693433
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115579
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928908
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.45851799
0 to x2 (h4 by hand)=  5.45851765
x2 to x3 (h5 computationally)=  1.21906885
x2 to x3 (h5 by hand)=  1.21907017
x2 to x3 (h5 by handdddd)=  1.21906933
x3 to x1 (h6 computationally)=  3.5738606
x3 to x1 (h6 by hand)=  3.5738613
K22  (computationally) = 0.241731692
K22  (by hand) = 0.241731602
K32  (computationally) = 0.226075399
K32  (by hand) = 0.226075335
K21  (computationally) = 0.259153707
K21  (by hand) =0.259153698
K31  (computationally) = 0.257498067
K31  (by hand) = 0.257498408
K33  (computationally) = 0.214665863
K33  (by hand) = 0.214666783
K11  (computationally) = 0.445272067
K11  (by hand) = 0.445273518
x2= 0.138717575
x3= 0.156453691
x1=  0.294878943
T by hand=816.568793
V by hand=-190.016401
m2=34.1
R=  0.0328721919
Energywithoutmass = 1137.94562
Energy=  25281.9456
R=  0.0328721919
```

```
normx2 (computationally)=  -0.606693442
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115582
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928904
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.47485782
0 to x2 (h4 by hand)=  5.47485754
x2 to x3 (h5 computationally)=  1.22278907
x2 to x3 (h5 by hand)=  1.22279034
x2 to x3 (h5 by handdddd)=  1.22278955
x3 to x1 (h6 computationally)=  3.58512617
x3 to x1 (h6 by hand)=  3.58512679
K22  (computationally) = 0.242091161
K22  (by hand) = 0.242091083
K32  (computationally) = 0.226412323
K32  (by hand) = 0.226412267
K21  (computationally) = 0.25955069
K21  (by hand) =0.259550682
K31  (computationally) = 0.257892654
K31  (by hand) = 0.257892986
K33  (computationally) = 0.214986324
K33  (by hand) = 0.214987224
K11  (computationally) = 0.445960866
K11  (by hand) = 0.445962175
x2= 0.138510333
x3= 0.156218602
x1=  0.294418105
T by hand=819.105902
V by hand=-190.306555
m2=34.2
R=  0.0328721919
Energywithoutmass = 1137.79869
Energy=  25281.7987
R=  0.0328721919
normx2 (computationally)=  -0.606693411
```

```
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115591
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928907
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.49119778
0 to x2 (h4 by hand)=  5.49119733
x2 to x3 (h5 computationally)=  1.22651018
x2 to x3 (h5 by hand)=  1.22651185
x2 to x3 (h5 by handdddd)=  1.22651067
x3 to x1 (h6 computationally)=  3.59639291
x3 to x1 (h6 by hand)=  3.59639351
K22  (computationally) = 0.242449992
K22  (by hand) = 0.242449866
K32  (computationally) = 0.226748705
K32  (by hand) = 0.226748616
K21  (computationally) = 0.25994709
K21  (by hand) =0.259947078
K31  (computationally) = 0.258286825
K31  (by hand) = 0.258287257
K33  (computationally) = 0.215306749
K33  (by hand) = 0.215307914
K11  (computationally) = 0.44664885
K11  (by hand) = 0.446650141
x2= 0.138304008
x3= 0.155984567
x1=  0.293959384
T by hand=821.643264
V by hand=-190.596401
m2=34.3
R=  0.0328721919
Energywithoutmass = 1137.67076
Energy=  25 281.6708
R=  0.0328721919
normx2 (computationally)=  -0.606693468
normx2 (by hand)=  -0.606693562
```

```
normx3 (computationally)=  -0.548115597
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928906
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.50753901
0 to x2 (h4 by hand)=  5.50753886
x2 to x3 (h5 computationally)=  1.23023013
x2 to x3 (h5 by hand)=  1.23023134
x2 to x3 (h5 by handdddd)=  1.23023062
x3 to x1 (h6 computationally)=  3.60766074
x3 to x1 (h6 by hand)=  3.60766129
K22  (computationally) = 0.242808546
K22  (by hand) = 0.242808506
K32  (computationally) = 0.227084711
K32  (by hand) = 0.227084683
K21  (computationally) = 0.260342918
K21  (by hand) =0.260342914
K31  (computationally) = 0.258680105
K31  (by hand) = 0.258680422
K33  (computationally) = 0.215625868
K33  (by hand) = 0.21562674
K11  (computationally) = 0.447335665
K11  (by hand) = 0.447336827
x2= 0.138098604
x3= 0.155751561
x1=  0.293502757
T by hand=824.180756
V by hand=-190.885648
m2=34.4
R=  0.0328721919
Energywithoutmass = 1137.56175
Energy=  25 281.5618
R=  0.0328721919
normx2 (computationally)=  -0.606693504
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115579
```

```
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928907
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.52388051
0 to x2 (h4 by hand)=  5.52388056
x2 to x3 (h5 computationally)=  1.23395016
x2 to x3 (h5 by hand)=  1.23395084
x2 to x3 (h5 by handdddd)=  1.23395066
x3 to x1 (h6 computationally)=  3.61892996
x3 to x1 (h6 by hand)=  3.61893066
K22  (computationally) = 0.243166519
K22  (by hand) = 0.243166533
K32  (computationally) = 0.227420182
K32  (by hand) = 0.227420191
K21  (computationally) = 0.260738176
K21  (by hand) =0.260738178
K31  (computationally) = 0.259072789
K31  (by hand) = 0.259072969
K33  (computationally) = 0.215944397
K33  (by hand) = 0.215944903
K11  (computationally) = 0.448021984
K11  (by hand) = 0.448023382
x2= 0.137894104
x3= 0.155519585
x1=  0.293048218
T by hand=826.718506
V by hand=-191.174564
m2=34.5
R=  0.0328721919
Energywithoutmass = 1137.47138
Energy=  25 281.4714
R=  0.0328721919
normx2 (computationally)=  -0.606693504
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115579
normx3 (by hand)=  -0.54811561
```

```
normx1 (computationally)=  -0.434928905
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.54022217
0 to x2 (h4 by hand)=  5.54022223
x2 to x3 (h5 computationally)=  1.23767125
x2 to x3 (h5 by hand)=  1.23767193
x2 to x3 (h5 by handdddd)=  1.23767175
x3 to x1 (h6 computationally)=  3.63019999
x3 to x1 (h6 by hand)=  3.63020067
K22  (computationally) = 0.243523877
K22  (by hand) = 0.243523892
K32  (computationally) = 0.227755134
K32  (by hand) = 0.227755145
K21  (computationally) = 0.261132845
K21  (by hand) =0.261132847
K31  (computationally) = 0.259465088
K31  (by hand) = 0.259465268
K33  (computationally) = 0.216263022
K33  (by hand) = 0.216263527
K11  (computationally) = 0.448706885
K11  (by hand) = 0.448708228
x2= 0.137690501
x3= 0.155288642
x1=  0.292595742
T by hand=829.256431
V by hand=-191.463058
m2=34.6
R=  0.0328721919
Energywithoutmass = 1137.39943
Energy=  25 281.3994
R=  0.0328721919
normx2 (computationally)=  -0.60669348
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.5481156
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928904
```

```
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.55656409
0 to x2 (h4 by hand)=  5.55656401
x2 to x3 (h5 computationally)=  1.24139331
x2 to x3 (h5 by hand)=  1.24139446
x2 to x3 (h5 by handdddd)=  1.24139381
x3 to x1 (h6 computationally)=  3.6414709
x3 to x1 (h6 by hand)=  3.64147141
K22  (computationally) = 0.24388065
K22  (by hand) = 0.243880629
K32  (computationally) = 0.228089588
K32  (by hand) = 0.228089573
K21  (computationally) = 0.261526932
K21  (by hand) =0.26152693
K31  (computationally) = 0.259856989
K31  (by hand) = 0.259857291
K33  (computationally) = 0.216581694
K33  (by hand) = 0.216582528
K11  (computationally) = 0.449390508
K11  (by hand) = 0.449391587
x2= 0.13748779
x3= 0.155058725
x1=  0.292145315
T by hand=831.794548
V by hand=-191.751157
m2=34.7
R=  0.0328721919
Energywithoutmass = 1137.34568
Energy=  25 281.3457
R=  0.0328721919
normx2 (computationally)=  -0.606693461
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115591
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928904
normx1 (by hand)=  -0.434928918
```

```
0 to x2 (h4 computationally)=  5.57290651
0 to x2 (h4 by hand)=  5.57290633
x2 to x3 (h5 computationally)=  1.24511493
x2 to x3 (h5 by hand)=  1.24511616
x2 to x3 (h5 by handdddd)=  1.24511543
x3 to x1 (h6 computationally)=  3.65274324
x3 to x1 (h6 by hand)=  3.65274382
K22  (computationally) = 0.244236922
K22  (by hand) = 0.244236872
K32  (computationally) = 0.228423541
K32  (by hand) = 0.228423506
K21  (computationally) = 0.261920462
K21  (by hand) =0.261920457
K31  (computationally) = 0.260248187
K31  (by hand) = 0.260248507
K33  (computationally) = 0.216899461
K33  (by hand) = 0.216900339
K11  (computationally) = 0.450073687
K11  (by hand) = 0.450074897
x2= 0.137285966
x3= 0.15482981
x1=  0.291696929
T by hand=834.332905
V by hand=-192.038885
m2=34.8
R=  0.0328721919
Energywithoutmass = 1137.31004
Energy=  25 281.31
R=  0.0328721919
normx2 (computationally)=  -0.606693467
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115579
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928904
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.58924962
```

```
0 to x2 (h4 by hand)=  5.58924947
x2 to x3 (h5 computationally)=  1.24883633
x2 to x3 (h5 by hand)=  1.24883737
x2 to x3 (h5 by handdddd)=  1.24883683
x3 to x1 (h6 computationally)=  3.66401671
x3 to x1 (h6 by hand)=  3.66401738
K22  (computationally) = 0.244592751
K22  (by hand) = 0.244592709
K32  (computationally) = 0.228757044
K32  (by hand) = 0.228757015
K21  (computationally) = 0.262313428
K21  (by hand) =0.262313424
K31  (computationally) = 0.260638722
K31  (by hand) = 0.260638994
K33  (computationally) = 0.217216501
K33  (by hand) = 0.217217245
K11  (computationally) = 0.450755928
K11  (by hand) = 0.450757302
x2= 0.137085025
x3= 0.154601895
x1=  0.291250563
T by hand=836.871449
V by hand=-192.326165
m2=34.9
R=  0.0328721919
Energywithoutmass = 1137.29231
Energy=  25 281.2923
R=  0.0328721919
normx2 (computationally)=  -0.606693467
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115598
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928904
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.60559315
0 to x2 (h4 by hand)=  5.605593
```

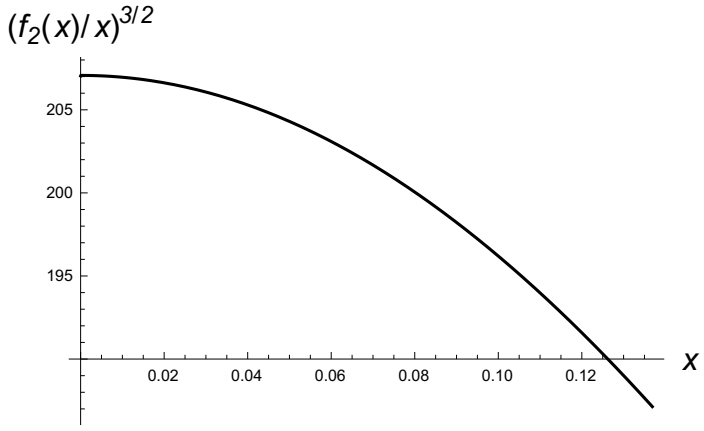
```
x2 to x3 (h5 computationally)=  1.25255859
x2 to x3 (h5 by hand)=  1.25255986
x2 to x3 (h5 by handdddd)=  1.2525591
x3 to x1 (h6 computationally)=  3.67529088
x3 to x1 (h6 by hand)=  3.67529141
K22  (computationally) = 0.244948056
K22  (by hand) = 0.244948014
K32  (computationally) = 0.229090089
K32  (by hand) = 0.229090059
K21  (computationally) = 0.262705816
K21  (by hand) =0.262705812
K31  (computationally) = 0.261028842
K31  (by hand) = 0.261029173
K33  (computationally) = 0.217533561
K33  (by hand) = 0.217534471
K11  (computationally) = 0.451436627
K11  (by hand) = 0.451437761
x2= 0.136884957
x3= 0.154374982
x1=  0.290806198
T by hand=839.410144
V by hand=-192.612988
m2=35.
R=  0.0328721919
Energywithoutmass = 1137.29224
Energy=  25 281.2922
R=  0.0328721919
normx2 (computationally)=  -0.606693468
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115578
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928904
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.62193714
0 to x2 (h4 by hand)=  5.62193699
x2 to x3 (h5 computationally)=  1.25628032
```

```
x2 to x3 (h5 by hand)=  1.25628136
x2 to x3 (h5 by handdddd)=  1.25628083
x3 to x1 (h6 computationally)=  3.68656656
x3 to x1 (h6 by hand)=  3.68656725
K22  (computationally) = 0.245302854
K22  (by hand) = 0.245302813
K32  (computationally) = 0.229422626
K32  (by hand) = 0.229422597
K21  (computationally) = 0.263097656
K21  (by hand) =0.263097652
K31  (computationally) = 0.261418243
K31  (by hand) = 0.261418514
K33  (computationally) = 0.21784964
K33  (by hand) = 0.217850383
K11  (computationally) = 0.452117036
K11  (by hand) = 0.452118429
x2= 0.136685757
x3= 0.154149049
x1=  0.29036383
T by hand=841.94909
V by hand=-192.899463
m2=35.1
R=  0.0328721919
Energywithoutmass = 1137.30973
Energy=  25 281.3097
R=  0.0328721919
normx2 (computationally)=  -0.606693467
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115597
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928905
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.60722753
0 to x2 (h4 by hand)=  5.60722738
x2 to x3 (h5 computationally)=  1.25293078
x2 to x3 (h5 by hand)=  1.25293203
```

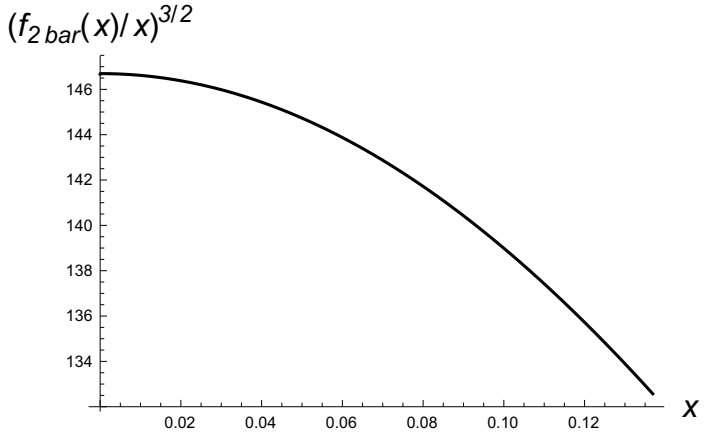
```
x2 to x3 (h5 by handdddd)=  1.25293128
x3 to x1 (h6 computationally)=  3.67641839
x3 to x1 (h6 by hand)=  3.67641893
K22  (computationally) = 0.244983558
K22  (by hand) = 0.244983517
K32  (computationally) = 0.229123364
K32  (by hand) = 0.229123335
K21  (computationally) = 0.262745025
K21  (by hand) =0.262745021
K31  (computationally) = 0.261067811
K31  (by hand) = 0.26106814
K33  (computationally) = 0.217565204
K33  (by hand) = 0.217566106
K11  (computationally) = 0.451504699
K11  (by hand) = 0.451505848
x2= 0.136864998
x3= 0.154352345
x1=  0.290761872
T by hand=839.664029
V by hand=-192.641654
m2=35.01
R=  0.0328721919
Energywithoutmass = 1137.2932
Energy=  25 281.2932
R=  0.0328721919
normx2 (computationally)=  -0.606693467
normx2 (by hand)=  -0.606693562
normx3 (computationally)=  -0.548115589
normx3 (by hand)=  -0.54811561
normx1 (computationally)=  -0.434928918
normx1 (by hand)=  -0.434928918
0 to x2 (h4 computationally)=  5.60575659
0 to x2 (h4 by hand)=  5.60575643
x2 to x3 (h5 computationally)=  1.25259565
x2 to x3 (h5 by hand)=  1.25259681
x2 to x3 (h5 by handdddd)=  1.25259615
```

```
x3 to x1 (h6 computationally)=  3.67540403
x3 to x1 (h6 by hand)=  3.67540482
K22  (computationally) = 0.244951606
K22  (by hand) = 0.244951565
K32  (computationally) = 0.229093409
K32  (by hand) = 0.22909338
K21  (computationally) = 0.262709754
K21  (by hand) =0.26270975
K31  (computationally) = 0.261032718
K31  (by hand) = 0.261033022
K33  (computationally) = 0.217536608
K33  (by hand) = 0.217537442
K11  (computationally) = 0.451444143
K11  (by hand) = 0.451445744
x2= 0.136882961
x3= 0.154372716
x1=  0.290801773
T by hand=839.435621
V by hand=-192.616
m2=35.001
R=  0.0328721919
Energywithoutmass = 1137.29232
Energy=  25 281.2923
x1=0.290801773
x3=0.154372716
x2=0.136882961
Kinetic Energy=839.435621
Potential Energy=-192.616
Volume Energy=490.472698
```

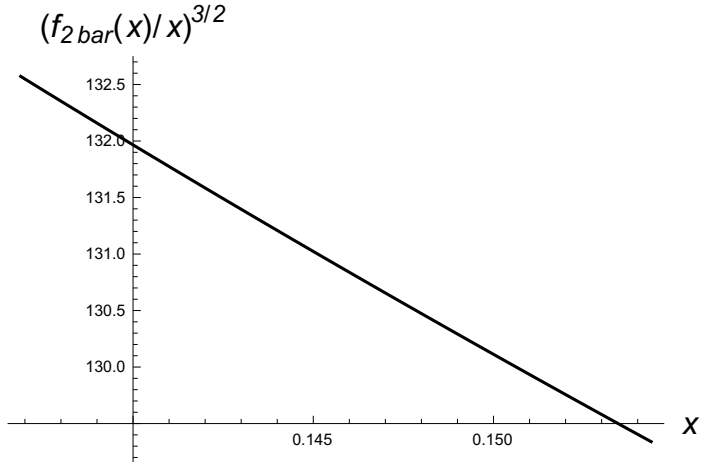
Out[ ]=


$(f_2(x)/x)^{3/2}$
205
200
195
0.02
0.04
0.06
0.08
0.10
0.12
x

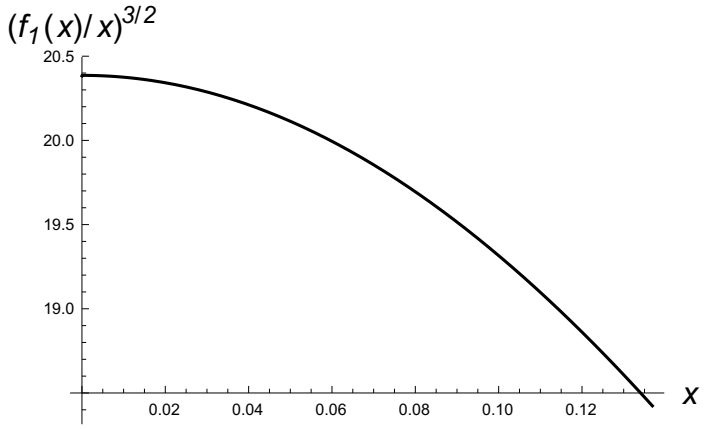
Out[ ]=


$(f_{2\,bar}(x)/x)^{3/2}$
146
144
142
140
138
136
134
0.02
0.04
0.06
0.08
0.10
0.12
x

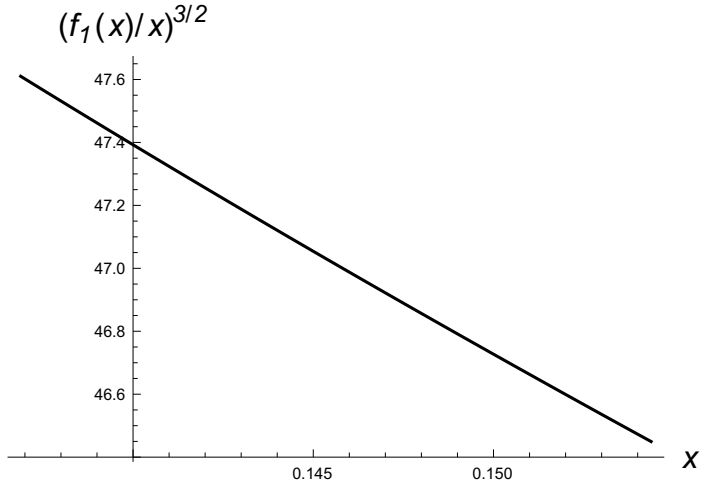
Out[ ]=


$(f_{2\,bar}(x)/x)^{3/2}$
132.5
132.0
131.5
131.0
130.5
130.0
0.145
0.150
x

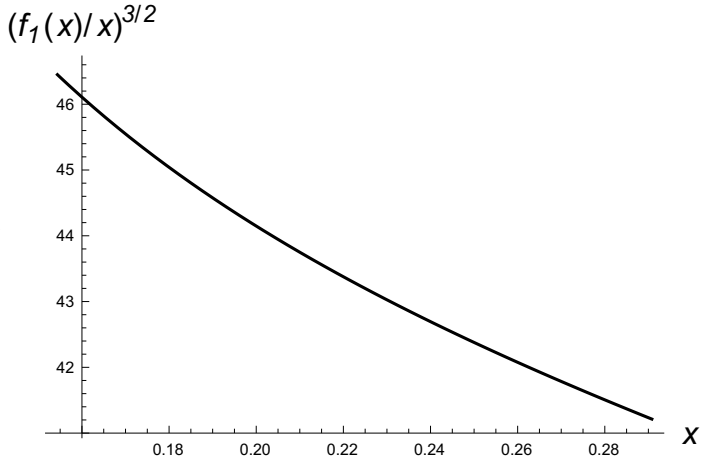
Out[ ]=


$(f_1(x)/x)^{3/2}$
20.5
20.0
19.5
19.0
0.02
0.04
0.06
0.08
0.10
0.12
x

Out[ ]=

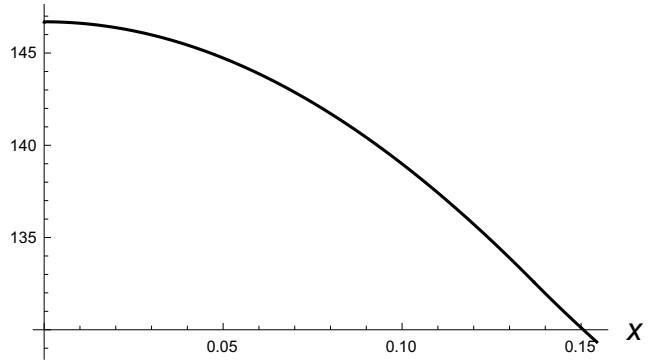

$(f_1(x)/x)^{3/2}$
47.6
47.4
47.2
47.0
46.8
46.6
0.145
0.150
x

Out[ ]=


$(f_1(x)/x)^{3/2}$
46
45
44
43
42
0.18
0.20
0.22
0.24
0.26
0.28
x

$(f_{2\,bar}(x)/x)^{3/2}$

Out[ ]=




$(f_1(x)/x)^{3/2}$

Out[ ]=

